HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

PHAN MINH HÒA
hoa.pm225495@sis.hust.edu.vn

Program: Data Science and Artificial Intelligence

Supervisor: Trịnh Văn Chiến

Department: Faculty of Computer Science

School: School of Information and Communications Technology

HANOI, 06/2026

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

PHAN MINH HÒA
hoa.pm225495@sis.hust.edu.vn

Program: Data Science and Artificial Intelligence

Supervisor: Trịnh Văn Chiến ______________

Signature

Department: Faculty of Computer Science

School: School of Information and Communications Technology

HANOI, 06/2026

ACKNOWLEDGMENT

I am grateful to my supervisor, **Trịnh Văn Chiến**, for the guidance, careful feedback, and patience that shaped this thesis. I thank the lecturers and staff of the School of Information and Communications Technology at Hanoi University of Science and Technology for the foundation they gave me, and the creators of the CMOSE [1] and DAiSEE [2] datasets, whose work made this study possible. Finally, I thank my family and friends for their constant support, and acknowledge my own persistence in seeing this work through.

ABSTRACT

Reading student engagement from video automatically is a useful tool for online and hybrid classes, yet it is hard: engagement labels are *ordinal* (Highly-Disengage, Disengage, Engage, Highly-Engage), badly *imbalanced*, and subjective, and models are rarely tested on any dataset other than the one they were trained on. This thesis studies engagement classification from per-frame facial-behaviour features (OpenFace) and clip-level video-motion features (I3D), and makes three contributions: a semantic-group hybrid architecture, a self-collected private test set, and an honest cross-dataset study. A study of single-encoder (monolithic) baselines first fixes how models are measured before any model is compared: the six candidate metrics clearly disagree on the best model — a disagreement the cross-dataset study later settles against accuracy — so quadratic-weighted kappa (QWK), macro-accuracy, and macro-MAE are chosen as the main metrics; the loss is shown to be a trade-off within those metrics, so cross-entropy is used for development; and a clip-level error analysis finds the OpenFace and I3D feature families make different errors, which is why they are combined.

First, the thesis proposes a **semantic-group hybrid temporal architecture**. Instead of feeding all 709 OpenFace features into one model, the features are split into five meaningful groups — gaze, eye landmarks, face landmarks, head pose, and action units — and each group is given its own temporal encoder (a Temporal Convolutional Network, a Transformer, or an LSTM), optionally combined with an I3D motion stream and trained with a small head on each stream. A full ablation over all 243 per-group encoder choices, with and without I3D, shows that the I3D hybrid family sits above the best monolithic baseline on the CMOSE test set (median quadratic-weighted kappa, QWK, of 0.553 against 0.537; best 0.605), that the strongest configurations mix encoder types, and that the only group with a clear encoder preference is **head pose, which prefers a TCN**.

Second, it introduces a **self-collected, hand-labelled private test set** of webcam clips, made with the same OpenFace/I3D pipeline as the public datasets and kept only for testing, as a test of real-world generalisation on a group of people who show engagement differently from the public datasets.

Third, it runs a **3$\times$3 cross-dataset study** (training on CMOSE, DAiSEE, or both together; testing on each public test set plus the private set) and treats the private set as the deciding real-world test. Engagement models do **not** transfer across datasets — off-diagonal QWK drops to near zero even when plain accuracy looks fine —

and the three contributions *compound on the unseen private set*: training on both datasets together with the semantic-group hybrid gives the best private-set result (QWK 0.379 against 0.285 for the best baseline, a bigger gap than in-domain). The thesis argues that order-aware, class-balanced metrics — quadratic-weighted kappa, macro-accuracy, and macro-MAE — not plain accuracy, must be the main measure for imbalanced engagement data.

Keywords: student engagement recognition, affective computing, OpenFace, I3D, temporal convolutional networks, ordinal classification, class imbalance, cross-dataset generalisation.

Student

(Signature and full name)

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

Abbreviation	Definition
AMP	Automatic Mixed Precision
AU	Facial Action Unit
BiLSTM	Bidirectional Long Short-Term Memory network
CDF	Cumulative Distribution Function
CE	Cross-Entropy loss
CMOSE	Class-level Multimodal Online Student Engagement dataset
CNN	Convolutional Neural Network
DAiSEE	Dataset for Affective States in E-Environments
EMD	Earth Mover’s Distance (ordinal loss)
GELU	Gaussian Error Linear Unit
I3D	Inflated 3D ConvNet (spatiotemporal video feature extractor)
LSTM	Long Short-Term Memory network
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
PCA	Principal Component Analysis
QWK	Quadratic-Weighted Kappa
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SMOTE	Synthetic Minority Over-sampling Technique
SOTA	State Of The Art
SVD	Singular Value Decomposition
TCN	Temporal Convolutional Network

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Online and hybrid education has become a permanent part of modern teaching. Lectures, tutorials, and exams are now usually given through video calls and learning platforms, and much learning happens with a webcam as the only link between the student and the teacher. This change brings both a chance and a problem. The chance is that the webcam stream shows a lot about how a student is following the material; the problem is that, unlike in a real classroom, the teacher cannot read the room. A teacher facing forty live video tiles has no real way of noticing that one student quietly lost focus ten minutes ago. *Automatic engagement recognition* aims to close this gap: it turns a short video of one student into an estimate of how engaged that student is, so that teachers, tutoring systems, and learning platforms can step in, change the pace of a lesson, or measure the learning experience at a scale that watching by hand can never reach.

This thesis studies the exact form of that problem used by the research community and by the datasets we build on. The input is a short video clip of one student, about ten seconds long, recorded from a front webcam. The output is one of four *ordinal* engagement levels: **Highly-Disengage (HD)**, **Disengage (DE)**, **Engage (EG)**, and **Highly-Engage (HE)**. These four levels form a scale from full withdrawal to active, focused participation, so the task is a four-class ordinal classification problem, not an unordered, nominal one.

Although it looks like ordinary video classification, engagement recognition is harder than a typical image- or action-recognition task for three reasons that come back throughout this thesis and shape every design choice in it.

1. **The labels are ordinal, not nominal.** The four classes lie on a line: HD $<$ DE $<$ EG $<$ HE. Predicting *Highly-Engage* for a *Highly-Disengage* student is a serious error, while predicting *Engage* for a *Highly-Engage* student is almost right. Plain accuracy ignores this order, treating both mistakes as equally wrong. So accuracy is the wrong training goal and the wrong evaluation metric; the size of an error must be weighted by how far apart the true and predicted classes are on the scale.

2. **The labels are badly imbalanced.** Real classes are mostly attentive, so engagement datasets are dominated by the middle and upper levels. In the CMOSE dataset about 69% of clips are labelled *Engage* and fewer than 3% are *Highly-Disengage*

(Section 3.2, Table 3.1). So a model can reach high accuracy by always guessing the majority class and never predicting disengagement at all — exactly the rare but important case that a useful monitoring system exists to catch.

3. **Engagement is subjective and dataset-specific.** Engagement is not a physical quantity with a true value; it is a human judgement made under a particular rubric, recording setup, and group of people. Different datasets differ in all three, so a model trained on one dataset meets data it has never seen the moment it is deployed. The group of people is not a small part of this change: students from different cultural and ethnic groups show the same inner engagement differently on the outside — display rules, gaze and head-movement habits, and baseline facial motion all vary across groups — so a model tuned on one group's style can misread another's in a regular way. How well models survive this change is rarely measured, even though deployment always means unseen data, usually from a group of people different from the one the model was trained on.

These three properties together are what make the problem both practically important and technically interesting. A system that is accurate on paper but blind to disengagement, or that works on the dataset it was trained on but falls apart on real video, is of no use in a classroom. So the main concern of this thesis is not headline accuracy but *ordinal agreement on imbalanced data, and how well it holds under distribution shift*.

These properties are not abstract: they follow directly from how an engagement signal would actually be used. The real unit of decision is rarely a single clip but a summary — an engagement curve over one session for one student, or a class-level spread the teacher watches in real time — and the value of such a signal lies almost fully in the tails: confirming that an already-attentive class is attentive tells you little, while flagging the few moments when a student slips into disengagement is the whole point, exactly the rare, extreme case that imbalance hides and accuracy fails to reward. A deployment also never sees the data it was tuned on — cameras, lighting, who is in the class, and the labelling rubric all change — so an in-domain number means nothing unless it is shown to survive that change, which is why this thesis treats a self-collected, never-trained-on test set and a clear cross-dataset protocol as the setting in which any claim of usefulness must finally be judged.

1.2 Background and Problems of Research

Research on vision-based engagement recognition has, broadly, taken two routes. The first encodes each video frame with a facial-behaviour toolkit — most often

OpenFace [3], which estimates gaze direction, head pose, facial landmarks, and action-unit intensities — and feeds the resulting feature sequence to a classical classifier or a recurrent network, for example an OpenFace+BiLSTM pipeline [4] or classical models such as logistic regression, support vector machines, and gradient boosting on summary features [5]. The second route learns straight from pixels with end-to-end deep video models, for instance an EfficientNetV2 backbone followed by an LSTM [6], or borrows context- and relationship-modelling ideas from affect recognition [7]. The two large datasets used in this thesis sit at the centre of this work: CMOSE [1] is a recent, large, expert-labelled multimodal dataset and the only one of our sources that comes with ready OpenFace and I3D features, and DAiSEE [2] is a widely used public dataset recorded "in the wild" that gives only raw video, from which we make the same features ourselves. A systematic review of computer-vision engagement detection [8] lists these pipelines and confirms that class imbalance and label subjectivity are common, unsolved problems across the field.

Two specific limits of the existing work motivate this thesis.

First, facial-behaviour features are mixed but are encoded as one block. A single OpenFace frame in CMOSE is a 709-dimensional vector that bundles together signals of very different physical kinds: a few gaze direction values, hundreds of eye- and face-landmark coordinates, head-pose shifts and rotations, and action-unit strengths. These parts move at different speeds — the eyes flick fast and noisily, head turns are slow and smooth, action units fire in short bursts. The usual practice of joining all 709 numbers and pushing them through one sequence encoder forces a single time model onto signals that do not share one. It also makes the model harder to read, because the role of each behavioural channel is no longer separate.

Second, cross-dataset generalisation is almost never measured. The engagement literature reports in-domain numbers — train and test on the same dataset — almost only. Yet the only deployment that matters is on data the model has never seen, from a different group of people, camera, and lighting than the training dataset. Because engagement labels are subjective and the class balance differs sharply between datasets (CMOSE is *Engage*-dominated while DAiSEE is split between *Engage* and *Highly-Engage*), there is every reason to expect that models transfer poorly — but without measurement this stays a guess. One more danger makes it worse: on imbalanced data a model that simply predicts the majority class can post a decent cross-dataset *accuracy* while having learned nothing that transfers, so the very metric most papers report can hide the failure it should show.

These two limits define the gap this thesis addresses. The mixed-feature problem calls for a representation that keeps the structure of the facial features instead of flattening it; the generalisation problem calls for an evaluation protocol that measures transfer honestly, with metrics that cannot be fooled by majority prediction.

1.3 Research Objectives and Conceptual Framework

The goal of this thesis is to improve and, just as importantly, to *honestly describe* four-class ordinal engagement recognition from facial-behaviour and motion features. We do this through one architecture idea and one set of evaluation rules, written as four research questions.

- **RQ1 — Decomposition.** Does splitting the OpenFace features into semantic groups, each with its own temporal encoder, improve engagement classification over monolithic baselines that encode all 709 features together?
- **RQ2 — Encoder assignment.** If splitting helps, which group→encoder choices actually matter? Is there a behavioural channel whose timing really prefers one encoder family (convolutional, recurrent, or attention-based) over another?
- **RQ3 — Motion fusion.** Does adding a global I3D motion/appearance stream to the split model help, and does any gain hold across many configurations rather than come from a single lucky one?
- **RQ4 — Generalisation and losses.** How well do these models generalise across datasets, and how does the loss trade ordinal agreement against minority-class recall and balanced ordinal error?

Conceptual framework. The shared idea behind RQ1–RQ3, shown by the hybrid architecture in Chapter 3 (Figure 3.3), is to treat a clip as a *multi-stream temporal signal* instead of a single sequence. The OpenFace features are split into five meaningful groups — gaze, eye landmarks, face landmarks, head pose, and action units — and each group is encoded *on its own* into a fixed-length embedding by an encoder chosen to suit its timing. A global I3D clip embedding can be added as a sixth stream. The embeddings are joined and classified, and every stream also gets its own small head so that each behavioural channel has to be useful on its own. We call this "divide the face, then fuse" design the *semantic-group hybrid*, and it is the main object tested in this thesis against standard monolithic baselines, across datasets, and across loss functions. The rule behind RQ4 is to report ordinal agreement (quadratic-weighted kappa), balanced accuracy, and balanced ordinal error (macro-MAE) as the main metrics, never plain accuracy alone, and to test every model not only in-domain but on two unseen datasets, one of which we collected ourselves.

1.4 Contributions

This thesis makes three contributions.

1. **Architecture — a semantic-group hybrid that divides the face, then fuses.** We propose a temporal architecture for engagement classification that splits the 709-dimensional OpenFace features into five behavioural groups, encodes each group with its own chosen Temporal Convolutional Network (TCN), Transformer, or LSTM, optionally adds a global I3D motion stream, and trains every stream with a small head (Section 3.5). A full 243-configuration ablation of the per-group encoder choice, with and without the I3D stream, shows that the I3D hybrid family sits above the strongest monolithic baseline (median quadratic-weighted kappa, QWK, of 0.553 against 0.537, best 0.605), that the gain belongs to the family rather than to one lucky configuration, and that head pose is the only behavioural group with a clear encoder preference, preferring a TCN — a preference that also holds under distribution shift (Section 5.2).

2. **Dataset — a new, self-collected private test set.** We collect 366 webcam clips, cut to a fixed length, filtered for tracking quality, labelled by hand by the author through a purpose-built interface, and made with the same OpenFace/I3D pipeline as the public datasets. It is used only as a held-out, test-only check of real-world generalisation on a group of people who show engagement differently from the public datasets (Section 3.2).

3. **Generalization — an honest cross-dataset study measured by QWK, not accuracy.** We run a 3×3 cross-dataset study (training on CMOSE, DAiSEE, or both together; testing on each public test set plus the private set) showing that engagement models do not transfer out of the box, that accuracy hides this collapse while QWK shows it, and that training on the datasets pooled together is the simplest fix that works best. On the private set the architecture and the pooled-training idea *compound*: the hybrid trained on both datasets is the best model (QWK 0.379 against 0.285 for the best baseline — a bigger gap than in-domain), which shows the practical value of all three contributions together (Sections 4.5 and 5.6).

These contributions are backed by a reproducible analysis pipeline that turns about 1,500 trained runs into the figures and tables of this thesis, from feature extraction through training, cross-dataset evaluation, and figure/table generation.

1.5 Organization of Thesis

The remainder of this thesis is organised as follows.

Chapter 2 reviews the work on vision-based engagement recognition and introduces the building blocks the thesis relies on. It sets the scope of the study, looks at the related feature-based and end-to-end approaches with their strengths and weaknesses, and then presents the technical basics: the OpenFace and I3D feature extractors, the three temporal architecture families (TCN, LSTM, Transformer), the loss functions used for imbalanced ordinal labels, and the ordinal evaluation metrics.

Chapter 3 presents the methodology. It gives an overview of the full pipeline as a single diagram, describes the three training datasets and their class imbalance, the private test set, and the preprocessing that makes them comparable, sets out the monolithic baselines, and then gives the semantic-group decomposition and the proposed hybrid architecture built on top of it. It also fixes the loss functions and their exact forms, the training protocol, and the 3×3 cross-dataset evaluation protocol with its metrics.

Chapter 4 studies the *monolithic baselines* and fixes the experimental frame, in a protocol-first order: an evaluation-protocol section first fixes the main metrics (QWK, macro-accuracy, macro-MAE), the development dataset (CMOSE), and the development loss (cross-entropy), each justified from up-front reasoning and in-domain evidence; the chapter then reads the in-domain leaderboard, studies prediction agreement to show the OpenFace and I3D families make different errors, and finally takes the baselines out of domain in the 3×3 matrix, closing the loop on the protocol.

Chapter 5 introduces the *semantic-group hybrid* and tests it against that frame: it checks that the 486-configuration hybrid family repeats the protocol evidence of Chapter 4, answers the design questions (which encoder suits which group, and whether adding the I3D stream helps — an answer that depends on the test setting), and then takes the hybrid out of domain, where it wins every cell of the 3×3 matrix and compounds with dataset pooling to the best result on the self-collected private set (RQ4).

Chapter 6 sums up the findings, restates the contributions in the light of the evidence, discusses the limits of the study, and proposes concrete directions for future work, including going after the rarest disengaged class, explicit domain adaptation, and group encoders that are learned rather than tried one by one.

CHAPTER 2. LITERATURE REVIEW

This chapter sets the technical and scientific context for the thesis. It first fixes the scope of the study (Section 2.1), then analyses the related work and its limitations (Section 2.2). The remaining sections present the foundational knowledge the thesis builds on and that later chapters assume: the feature extractors that turn a video into numerical descriptors (Section 2.3), the three temporal architecture families used both as baselines and as the building blocks of the proposed model (Section 2.4), the loss functions for imbalanced ordinal targets (Section 2.5), the metrics used to evaluate ordinal imbalanced classification (Section 2.6), and the statistical measures used to compare and analyse the results (Section 2.7). The chapter closes by summarising how these pieces are combined in Chapter 3.

2.1 Scope of Research

This review is scoped to **vision-based automatic engagement recognition**: classifying a learner's engagement level from facial video, where the only input is a frontal webcam recording of a single person. Within this scope the thesis concentrates on four technical components, and the review is organised around them: (i) facial-behaviour and motion feature extraction, (ii) temporal sequence modelling of the per-frame descriptors, (iii) loss functions appropriate to imbalanced ordinal labels, and (iv) the metrics used to evaluate such models.

Several adjacent problems are deliberately left out of scope. We do not consider physiological-sensor engagement estimation (heart rate, electrodermal activity, eye-tracking hardware), nor interaction- or clickstream-based estimation from platform logs, because they require instrumentation beyond a webcam. We do not treat general facial-expression or emotion recognition as an end in itself, referring to it only where it informs the design of engagement features. Finally, the thesis works with OpenFace and I3D features at the descriptor level rather than re-training the extractors: CMOSE is the only corpus that ships these features precomputed, while for DAiSEE and the self-collected private set we extract them ourselves from the raw video using the same toolkits (Section 3.2); the internal training of those extractors is reviewed for understanding but is not re-implemented.

2.2 Related Work

Engagement datasets and benchmarks. DAiSEE [2] is the most widely used in-the-wild benchmark for engagement recognition. It provides short webcam clips annotated with four ordinal engagement levels and is recorded under varied, uncontrolled conditions, which makes it realistic but noisy. CMOSE [1] is a more recent, larger, expert-labelled multimodal corpus that ships precomputed OpenFace and I3D features — the only one of the three corpora used here to do so — and reports strong inter-rater reliability; its scale and label quality make it the primary dataset of this thesis. A systematic review of computer-vision engagement detection [8] surveys the common pipelines and explicitly identifies the field's two recurring difficulties — severe class imbalance and the subjectivity of engagement labels — as open problems, which is consistent with the motivation in Chapter 1.

Feature-based approaches. A large body of work first reduces each frame to a compact facial descriptor with a toolkit such as OpenFace [3] and then models the resulting sequence. Representative pipelines feed OpenFace action units, gaze, and head pose to a recurrent network, for example an OpenFace+BiLSTM model [4], or apply dimensionality reduction and resampling — principal component analysis or singular value decomposition together with synthetic minority oversampling — before a convolutional or fully-connected classifier [9]. Other studies aggregate the features over time and compare classical learners such as logistic regression, support vector machines, multilayer perceptrons, k-nearest neighbours, and gradient-boosted trees [5]. These works collectively establish that OpenFace descriptors carry a genuine engagement signal, but they share a structural assumption: the 709-dimensional descriptor is treated as a single homogeneous input vector, encoded by one model with one temporal inductive bias.

End-to-end video approaches. A second family learns directly from pixels. EfficientNetV2 [10] backbones followed by an LSTM [6] encode appearance per frame and integrate it over time, and context-aware affect-recognition models exploit spatial relationships within the scene [7]. These models can capture appearance and motion cues that landmark-based descriptors discard, but they are heavier to train, require the raw video rather than released features, and are less interpretable because the learned representation is not decomposed into nameable behavioural channels.

Gap. Two gaps persist across both families and define the contribution of this thesis. First, the *heterogeneous* facial descriptor is encoded *monolithically* rather than per behavioural subsystem, even though its constituent signals — gaze, eye and face geometry, head pose, action units — have markedly different temporal characteristics. Second, *cross-dataset generalisation is rarely measured*; in-domain numbers dominate the literature, and where accuracy is the headline metric, majority-class prediction can disguise a complete failure to transfer. The semantic-group hybrid proposed in Chapter 3 targets the first gap, and the 3×3 cross-dataset protocol of Chapter 5 targets the second.

2.3 Feature Extraction

2.3.1 OpenFace 2.0 facial-behaviour descriptors

OpenFace 2.0 [3] is an open-source facial-behaviour analysis toolkit that processes a video frame by frame and, for each frame, estimates four kinds of measurement: the two- and three-dimensional *facial landmarks* that outline the face and eye regions, the *head pose* (three translation and three rotation parameters) recovered from those landmarks, the *eye-gaze* direction vectors and angles for both eyes, and the intensity and presence of the *facial action units* of the Facial Action Coding System. Each frame is also tagged with a tracking-confidence and success flag, which downstream pipelines use to discard poorly tracked frames.

In CMOSE the per-frame descriptor used in this thesis is **709-dimensional**. A key observation, exploited throughout the thesis, is that these 709 numbers are not homogeneous: they decompose naturally into five behavioural groups by the meaning of their columns — **gaze** (8 dimensions), **eye landmarks** (280), **face landmarks** (340), **head pose** (46), and **action units** (35). These groups correspond to physically distinct subsystems with distinct dynamics: gaze and eye landmarks change rapidly and noisily; head pose drifts slowly and smoothly; action units fire in short, sparse bursts. Section 3.4 formalises this decomposition and Section 3.5 turns it into an architecture.

2.3.2 I3D motion and appearance features

The Inflated 3D ConvNet (I3D) [11] is a video representation obtained by "inflating" the two-dimensional convolutional filters of an ImageNet-pretrained image network into three dimensions, so that they operate over space and time jointly, and then pretraining the result on the large Kinetics action-recognition dataset. I3D thereby captures appearance and short-range motion that landmark-based descriptors omit. In the data used here, CMOSE ships a *single precomputed 1024-dimensional I3D embedding per clip*, and for DAiSEE and the private set we extract the same single 1024-dimensional embedding per clip from the raw video — in every case a globally pooled spatiotemporal descriptor of the whole clip rather than a per-frame sequence. This thesis therefore treats I3D as a compact, clip-level motion/appearance stream that complements the per-frame OpenFace descriptors, and is careful in its claims accordingly: the I3D stream contributes a global summary, not fine-grained temporal motion.

2.4 Temporal Architectures

A clip is a sequence of per-frame OpenFace descriptors, so a model must reduce a variable-length sequence to a single engagement prediction. Three architecture families dominate sequence modelling, and all three are used in this thesis — both as monolithic baselines (Section 3.3) and as the per-group encoders inside the hybrid (Section 3.5). They differ chiefly in how they let one time step influence another.

2.4.1 Temporal Convolutional Networks (TCN)

A Temporal Convolutional Network [12] applies one-dimensional convolutions along the time axis. Two design choices make it well suited to behavioural sequences. First, the convolutions are *causal*: the output at time t depends only on inputs at times $\leq t$, which is achieved by padding on the left and removing the surplus on the right. Second, successive layers use exponentially increasing *dilation*: a layer at depth l skips 2^l steps between the inputs it convolves, so a stack of L layers reaches a receptive field that grows as $\mathcal{O}(2^L)$ while keeping the number of parameters small. Each layer is wrapped in a residual block — two dilated convolutions with weight normalisation, a rectified-linear nonlinearity, and dropout, summed with a (possibly projected) copy of the input — which stabilises the training of deep stacks. The result is a strong inductive bias for *short, local temporal motifs* at multiple scales, which matches signals such as a head nod or an action-unit burst.

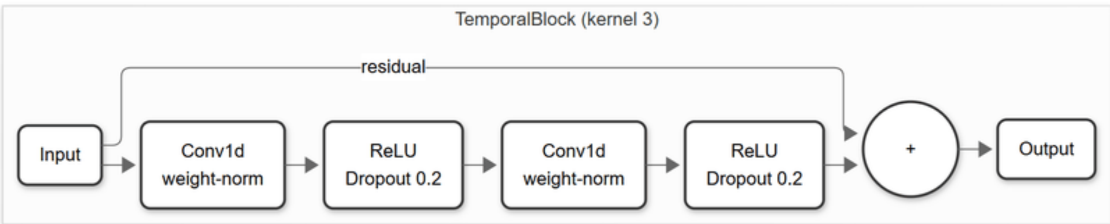


Figure 2.1: The residual TemporalBlock used in every TCN of this thesis: two weight-normalised dilated causal 1-D convolutions, each followed by a ReLU and dropout, summed with a residual connection (kernel size 3).

2.4.2 Long Short-Term Memory (LSTM)

The Long Short-Term Memory network [13] is a recurrent architecture that processes the sequence one step at a time while maintaining a cell state c_t and a hidden state h_t. Three gates — input i_t, forget f_t, and output o_t — regulate how much new information is written, how much past information is retained, and how much of the cell state is exposed:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t), \end{aligned} \tag{2.1}$$

where σ is the logistic sigmoid and $\odot$ is the element-wise product. The gating mechanism lets the network carry information across many time steps without the vanishing-gradient problem of plain recurrent networks, giving it a bias toward *longer-range dependencies*. The cost is that the sequential recurrence is harder to parallelise than convolution or attention.

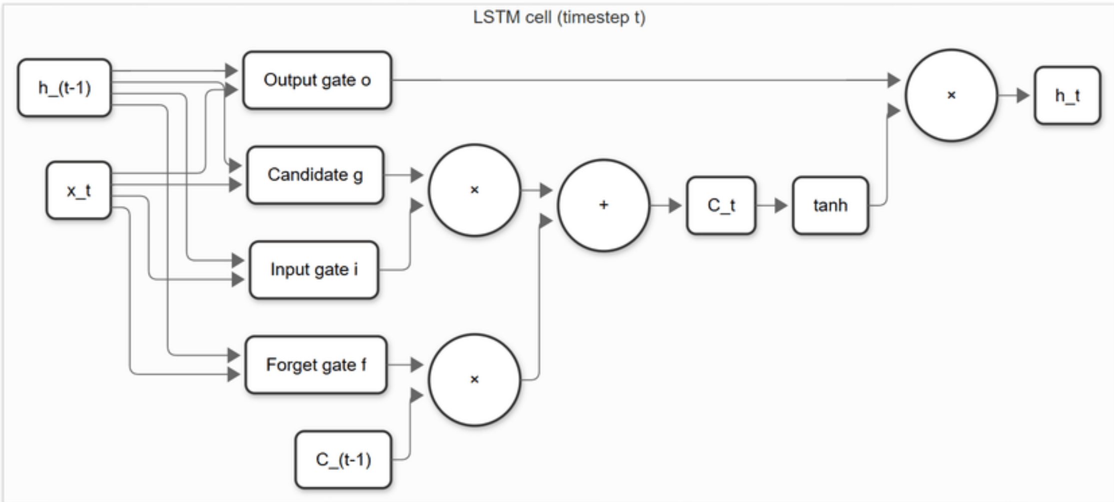


Figure 2.2: A single LSTM cell at time step t: the forget, input, and output gates regulate how the cell state C_t and hidden state h_t are updated from the previous states C_{t-1}, h_{t-1} and the input x_t.

2.4.3 Transformers

The Transformer encoder [14] dispenses with recurrence and convolution and instead relates all time steps directly through *self-attention*. For queries Q, keys K, and values V obtained by linear projections of the input, attention computes

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2.2}$$

so that every output position is a learned weighted combination of every input position, with d_k the key dimension. Because attention is permutation-invariant, order is injected through a *positional encoding* added to the input; this thesis uses the standard sinusoidal encoding. Multiple attention "heads" run in parallel and their outputs are concatenated, and each attention sub-layer is followed by a position-wise feed-forward network, with residual connections and layer normalisation throughout (Figure 2.3). The Transformer has the weakest temporal prior of the three families — it can attend anywhere — which makes it flexible but comparatively data-hungry.

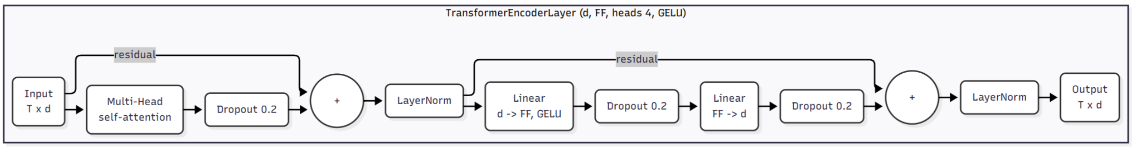


Figure 2.3: A single Transformer encoder layer used in this thesis: multi-head self-attention over the sinusoidally position-encoded input, followed by a position-wise feed-forward network, each sub-layer wrapped in a residual connection and layer normalisation.

2.5 Loss Functions for Imbalance and Ordinality

Let a model produce logits $z \in \mathbb{R}^C$ for the $C = 4$ engagement classes and softmax probabilities $\widehat{y}_c = \mathrm{softmax}(z)_c = e^{z_c} / \sum_j e^{z_j}$, and let $y \in \{1, \ldots, C\}$ be the ordinal ground-truth class with one-hot encoding $y_c = \mathbf{1}[\, c = y \,]$. Three loss families are compared in this thesis; their exact application (which weights, computed on which split, combined with which auxiliary term) is given in Section 3.6.

Cross-entropy (CE) is the standard classification loss,

$$\mathcal{L}_{\text{CE}} = -\sum_{c=1}^{C} y_c \log \widehat{y}_c, \tag{2.3}$$

which, because y is one-hot, reduces to the negative log-probability of the true class; it treats all classes symmetrically and all errors as equally severe. On imbalanced data it is dominated by the majority class and on ordinal data it ignores class order, so it serves as the baseline against which the other losses are judged.

Weighted cross-entropy counteracts imbalance by scaling each class's contribution by a weight w_c inversely proportional to its frequency,

$$\mathcal{L}_{\text{WCE}} = -\sum_{c=1}^{C} w_c \, y_c \log \widehat{y}_c. \tag{2.4}$$

This raises the cost of mistakes on rare classes such as *Highly-Disengage* and pushes the model away from collapsing onto the majority, at the expense of some headline accuracy. Resampling techniques such as SMOTE [15] address the same imbalance at the data level by synthesising minority examples; this thesis instead studies loss-level remedies, which require no change to the data pipeline.

Ordinal (EMD-style) loss encodes the order of the classes by comparing *cumulative* distributions. Writing the predicted cumulative distribution as $\sum_{j=1}^{k} \widehat{y}_j$ and the cumulative distribution of the one-hot label as the unit step $\sum_{j=1}^{k} y_j$ (which rises from 0 to 1 at the true class), the loss is the mean squared distance between the two step functions,

$$\mathcal{L}_{\text{EMD}} = \frac{w_y}{C} \sum_{k=1}^{C} \left(\sum_{j=1}^{k} \widehat{y}_j - \sum_{j=1}^{k} y_j \right)^2 , \tag{2.5}$$

a one-dimensional, squared Earth Mover's Distance between the predicted and target distributions over the ordered classes; the leading w_y is the weight of the true class (Section 3.6), set to 1 in the unweighted variant. Because moving probability mass across several classes costs more than moving it to an adjacent class, this loss directly penalises the large ordinal errors (HD↔HE) that accuracy and plain cross-entropy treat as ordinary.

All models are optimised with Adam [16], an adaptive first-order optimiser that maintains per-parameter running estimates of the first and second moments of the gradient and is the de-facto default for training such networks.

2.6 Evaluation Metrics

Because the labels are ordinal and imbalanced, the choice of metric is itself a research question (RQ4). This thesis reports six metrics over a test set of N samples with true labels y_i and predicted labels $\widehat{y}_i$ (for $i = 1, \ldots, N$), and treats three of them as primary. Where a confusion matrix is needed, O_{ij} denotes the number of samples of true class i predicted as class j.

Accuracy is the fraction of correct predictions,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\, y_i = \widehat{y}_i \,] . \tag{2.6}$$

It is reported for comparability with the literature but is explicitly *not* a primary metric here, because on imbalanced data it rewards majority prediction.

Macro-accuracy (balanced accuracy) is the mean of the per-class recalls,

$$\text{MacroAcc} = \frac{1}{C} \sum_{c=1}^{C} \frac{\sum_{i=1}^{N} \mathbf{1}[\, y_i = \widehat{y}_i = c \,]}{\sum_{i=1}^{N} \mathbf{1}[\, y_i = c \,]} , \tag{2.7}$$

giving every class equal weight regardless of frequency. It exposes the minority-class collapse that accuracy hides and is a **primary** metric.

Mean absolute error (MAE) treats the class indices as an ordinal scale and averages the distance between truth and prediction,

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \widehat{y}_i| , \tag{2.8}$$

while **macro-MAE** averages the per-class MAE so that errors on rare classes are not drowned out,

$$\mathrm{MacroMAE} = \frac{1}{C}\sum_{c=1}^{C} \frac{\sum_{i=1}^{N} \mathbf{1}[\,y_i = c\,]\,|y_i - \widehat{y}_i|}{\sum_{i=1}^{N} \mathbf{1}[\,y_i = c\,]}. \tag{2.9}$$

Both reward predictions that are at least *close* on the scale even when not exactly right; both are smaller-is-better. **Macro-MAE is a primary metric**: it is the natural ordinal-distance companion to QWK and macro-accuracy, reporting *how far* a prediction misses rather than only *whether* it misses, with each class weighted equally so the rare disengaged classes are not masked by the majority.

Cohen's kappa [17] measures agreement corrected for the agreement expected by chance. With observed agreement p_o and expected agreement p_e (computed from the marginals), $\kappa = (p_o - p_e)/(1 - p_e)$. In its weighted form it is written as

$$\kappa = 1 - \frac{\sum_{ij} w_{ij}\, O_{ij}}{\sum_{ij} w_{ij}\, E_{ij}}, \tag{2.10}$$

where E_{ij} is the expected count under independent marginals and w_{ij} is a penalty matrix. The unweighted variant uses $w_{ij} = 1$ off the diagonal and 0 on it.

Quadratic-weighted kappa (QWK) is the same agreement coefficient with the quadratic penalty

$$w_{ij} = \frac{(i-j)^2}{(C-1)^2}, \tag{2.11}$$

so that disagreements are penalised in proportion to the *square* of the ordinal distance between the true and predicted classes. QWK is therefore the natural single number for this problem: it is corrected for chance, it respects the class order, and it cannot be inflated by majority prediction. It is the **primary** metric of the thesis, with macro-accuracy as the primary balanced-recall companion and macro-MAE as the primary ordinal-distance companion.

2.7 Statistical Measures for Result Analysis

Beyond scoring each model in isolation, the results chapters ask three *comparative* questions that need their own statistical tools: do the six metrics *rank* the configurations the same way (Section 4.2.1), does in-domain strength *predict* cross-dataset strength (Section 4.5.3), and do different models make the *same per-clip decisions* (Section 4.4)? The first two are rank-correlation questions; the third is an agreement question.

Spearman's rank correlation ρ [18] measures how well a monotonic relationship describes two variables by applying the Pearson correlation coefficient to their ranks rather than their values. For n paired observations with rank differences d_i and no ties it reduces to

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n\left(n^2-1\right)}, \tag{2.12}$$

ranging from -1 (perfectly inverted ordering) through 0 (no monotonic relation) to $+1$ (identical ordering). This thesis uses ρ to test whether a model's in-domain QWK predicts its QWK on unseen target corpora.

Kendall's τ [19] measures rank agreement through *concordance* rather than squared rank distance. Counting the pairs of observations ordered the same way in both variables (concordant, n_c) against those ordered oppositely (discordant, n_d), and assuming no ties, the coefficient is

$$\tau_a = \frac{n_c - n_d}{\binom{n}{2}}, \tag{2.13}$$

again bounded by $[-1, 1]$. It is less sensitive to small samples and to a few large rank displacements than ρ, which is why it is used here to quantify how far the six evaluation metrics agree on the ranking of the baseline configurations.

Prediction agreement and ensemble headroom. To ask whether two models *agree clip by clip* rather than merely score alike, the thesis reuses Cohen's κ (Section 2.6) between the two models' predicted-label vectors; chance-correction matters because two majority-leaning models agree often by default. The headroom available from combining a pool of M configurations is bounded by the *oracle accuracy*, the fraction of clips that at least one configuration classifies correctly,

$$\mathrm{OracleAcc} = \frac{1}{N}\sum_{n=1}^{N} \mathbf{1}\left[\,\exists m \in \{1,\ldots,M\} : \widehat{y}_n^{(m)} = y_n\,\right], \tag{2.14}$$

which upper-bounds any fixed combiner, and is contrasted with the realistic *majority-vote accuracy* obtained by taking the modal prediction across the pool. A large gap between the best single model, the majority vote, and the oracle indicates that the models err on *different* clips and that an ensemble has room to improve.

2.8 Chapter Summary

This chapter scoped the thesis to webcam-based ordinal engagement recognition and analysed the two dominant approaches in the literature, identifying their shared limitations: heterogeneous facial descriptors are encoded monolithically, and cross-dataset generalisation is seldom measured with metrics that resist majority-class inflation. It then laid the technical foundations the rest of the thesis depends on

— the OpenFace 709-dimensional descriptor and its five behavioural groups, the clip-level I3D embedding, the three temporal architecture families (TCN, LSTM, Transformer) and their differing temporal biases, the three loss families for imbalance and ordinality, the six evaluation metrics with QWK, macro-accuracy, and macro-MAE as primary, and the rank-correlation (Spearman ρ, Kendall τ) and prediction-agreement statistics used to analyse the results. Chapter 3 combines these foundations into a concrete methodology: it turns the five-group decomposition into the proposed semantic-group hybrid architecture, specifies the baselines and the exact loss formulations, and defines the training and 3×3 cross-dataset evaluation protocols.

CHAPTER 3. METHODOLOGY

This chapter specifies the methodology of the thesis. Section 3.1 gives an overview of the complete pipeline as a single diagram. Section 3.2 describes the three training corpora, the self-collected private test set, and the preprocessing that makes them comparable. Section 3.3 defines the monolithic baseline models that anchor every comparison. Section 3.4 then formalises the semantic-group decomposition of the OpenFace descriptor, and Section 3.5 builds the proposed semantic-group hybrid architecture on top of it, in explicit contrast to those baselines. Section 3.6 fixes the exact loss formulations, Section 3.7 the training protocol, and Section 3.8 the 3×3 cross-dataset evaluation protocol and its metrics.

3.1 Overview

The task is framed as four-class ordinal classification of a fixed-length clip. Per-frame OpenFace descriptors and a clip-level I3D embedding are extracted, brought to a common temporal length, and normalised. The monolithic baselines (Section 3.3) follow the standard recipe: a single encoder consumes the full 709-dimensional OpenFace sequence (or the pooled I3D vector) and feeds a classifier. The proposed model replaces that single encoder with a split-and-encode design: the OpenFace descriptor is split into five semantic groups, each encoded independently into a 64-dimensional embedding by its own temporal encoder; in the multimodal variant the I3D clip vector is encoded into a 128-dimensional embedding by a small MLP. The embeddings are concatenated and classified by a shared multilayer perceptron, while an auxiliary head supervises every stream separately; the hybrid is shown in Figure 3.3. Every model is trained and evaluated under a 3×3 cross-dataset protocol and three loss functions.

3.2 Data and Preprocessing

3.2.1 Datasets

The thesis uses three training sources and three test sets, whose sizes and class balance are reported in Table 3.1.

- **CMOSE** [1]: 12,197 clips, divided into a `train` split (8,783), an `unlabel` split (2,193) that this thesis uses as the validation set for early stopping and checkpoint selection, and a `test` split (1,221).
- **DAiSEE** [2]: 8,571 clips, whose official `Train`/`Validation`/`Test` partitions are mapped onto the same `train`/`unlabel`/`test` roles (5,358 / 1,429 / 1,784).

- **Combined**: the union of CMOSE and DAiSEE (20,768 clips), with source-prefixed identifiers so the two corpora cannot collide. All three splits are merged split-wise, giving 14,141 / 3,622 / 3,005 train / validation / test clips.
- **Private (self-collected)**: a held-out set of **366 clips that we collected and manually labelled ourselves**, used **for testing only** — never in training or model selection. Its construction is described in Section 3.2.2.

All four sets share the four ordinal labels HD/DE/EG/HE. Table 3.1 and Figure 3.1 show that all three training corpora are heavily imbalanced toward the upper-middle classes, and — crucially — imbalanced in *different directions*. CMOSE is dominated by *Engage* (69.4%) with only 2.8% *Highly-Disengage*, whereas DAiSEE is split between *Engage* (50.3%) and *Highly-Engage* (43.8%) with just 0.7% *Highly-Disengage*. The rare disengaged classes are precisely the ones a useful monitoring system must catch, yet they amount to a fraction of a percent to a few percent of the data. The same imbalance holds within each of the train, validation, and test partitions, so it is a property of the problem rather than of a particular split.

Table 3.1: Dataset statistics and class balance (%).

Dataset	Total	Train	Val	Test	HD%	DE%	EG%	HE%
CMOSE	12197	8783	2193	1221	2.800	18.100	69.400	9.600
DaiSEE	8571	5358	1429	1784	0.700	5.100	50.300	43.800
Combined	20768	14141	3622	3005	2.000	12.800	61.500	23.700

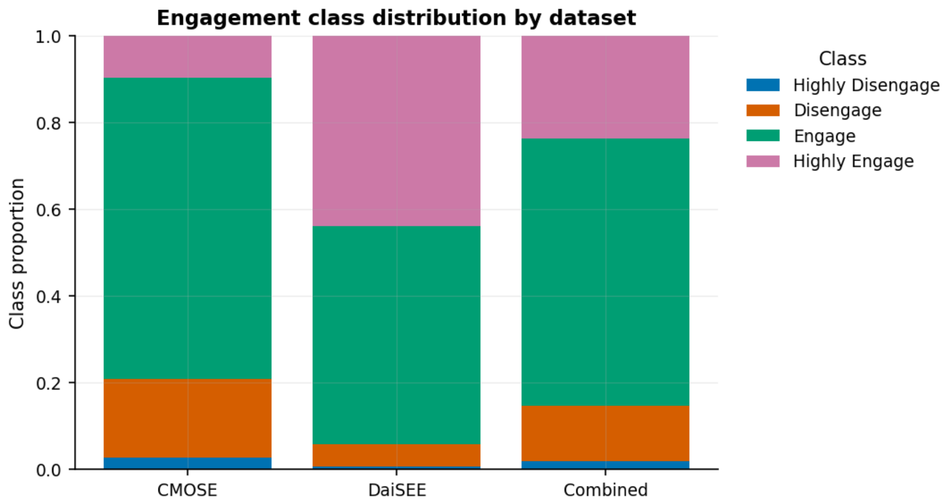


Figure 3.1: Engagement class distribution by dataset.

This severe and *heterogeneous* imbalance shapes three choices the thesis returns to. It predicts difficulty in transferring between corpora, where a model fit to one label prior meets a sharply different one at test time (Chapters 4 and 5); it motivates the imbalance-aware losses of Section 3.6; and, because predicting the majority class alone scores well on every corpus, it rules out raw accuracy in favour of quadratic-weighted kappa, macro-accuracy, and macro-MAE as the **primary** metrics (Section 3.8).

3.2.2 The private evaluation set

The private set measures deployment rather than in-corpus behaviour; because no model is ever trained or selected on it, it is the cleanest probe in the thesis. It also serves a purpose specific to engagement: because the way an inner state of engagement is outwardly displayed varies across cultural and ethnic groups, a model validated only on a foreign-population corpus offers no guarantee on a local deployment population; a small, locally collected, hand-labelled test set is the most direct way to check that a model is fit for its intended users before it is deployed. It was built in five steps.

1. **Source and segmentation.** Webcam recordings of online-class sessions were sampled at several points within each session and cut into fixed-length **10-second clips**, each containing a single learner.

2. **Tracking-quality filtering.** Every clip was run through the same OpenFace pipeline as the public corpora; clips with too few successfully tracked frames were rejected automatically, since a descriptor built from unreliable frames is meaningless.

3. **Manual labelling.** The accepted clips were labelled by the author through a purpose-built local interface, one engagement label per clip; reliance on a *single annotator* is a limitation revisited in Section 6.4.

4. **Feature extraction.** Accepted clips were processed through the same OpenFace (709-d per frame) and I3D (1024-d per clip) extraction pipeline we applied to the raw DAiSEE video — identical in format to the features CMOSE ships precomputed — so the private features are distributionally comparable to the training features.

5. **Final set.** Of the labelled clips, the **366** with complete, aligned OpenFace and I3D features form the private evaluation set. Its label prior differs again from both public corpora — 58.2% *Engage*, 30.6% *Highly-Engage*, 8.5% *Disengage*, and only 2.7% *Highly-Disengage* — which, together with its independent recording conditions, makes it the most realistic measure of how the models would behave in deployment. It anchors Section 5.6.

3.2.3 Features and preprocessing

Features. Each clip is represented by per-frame OpenFace 2.0 descriptors (709-d) [3] and a clip-level I3D embedding (1024-d) [11]. CMOSE is the only corpus that ships these features precomputed; for DAiSEE and the private set we extract both from the raw video with the same toolkits, so all three sources share an identical feature format. When a single OpenFace frame contains more than one detected face, the highest-confidence row is kept; non-numeric entries are coerced to zero.

Temporal resampling. OpenFace clips vary in length, so each is resampled along time to a fixed number of frames by linear interpolation per feature. The OpenFace sequence length is **300 frames** for the single-corpus runs and is reduced to **150** for the larger Combined runs to fit memory; in the multimodal variant the OpenFace stream is resampled to **75** frames. The I3D feature is a single 1024-dimensional vector per clip (not a sequence): the I3D stream mean-pools it to that vector and encodes it with an MLP (Section 3.5).

Normalisation. Every feature is standardised by a per-feature z-score whose mean and standard deviation are fit on the **training split only**; the validation, test, and private sets are transformed with those training statistics, so no target-set information leaks into preprocessing. OpenFace and I3D are standardised with separate statistics.

3.3 Baseline Models

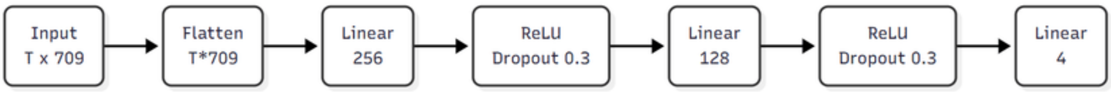
Five monolithic baselines provide the comparison points; they form the allow-list `MODEL_ORDER` used by the analysis. Each follows the standard recipe of the literature: a *single* encoder consumes the full feature sequence and feeds a classifier.

- `openface_mlp`: flattens the OpenFace sequence and classifies it with a multilayer perceptron (→256→128→4, dropout 0.3).
- `openface_tcn`: a TCN over the 709-d sequence (residual dilated blocks of widths $256, 128, 128$; kernel 3; dilations $1, 2, 4$), adaptive average-pooling, then a $128 \to 128 \to 4$ classifier.
- `openface_lstm`: a two-layer LSTM (hidden width 256) whose final hidden state feeds a $256 \to 128 \to 4$ classifier.
- `openface_transformer`: a linear projection to 128 dimensions, a sinusoidal positional encoding, and two Transformer encoder layers (4 heads, feed-forward width 256), mean-pooled and classified.

- `i3d_mlp`: mean-pools the I3D clip feature to a 1024-d vector and classifies it with an MLP ($1024 \to 256 \to 128 \to 4$, dropout 0.3).

A sixth model, `openface_tcn_i3d_fusion` (a shared-fusion TCN over both modalities), exists in the codebase but is **excluded from the assessment** so that the comparison isolates the semantic-group hypothesis rather than generic modality fusion.

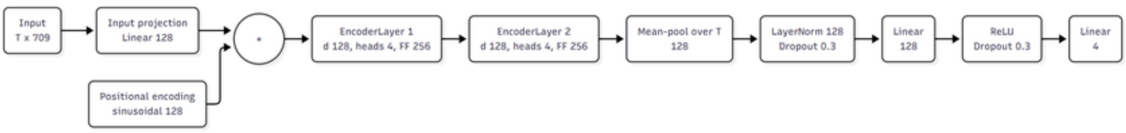
The five baseline architectures are shown in Figure 3.2. The four OpenFace baselines (a–d) encode the full $T \times 709$ sequence with a single encoder, while `i3d_mlp` (e) classifies the pooled 1024-d I3D clip vector. Forcing one encoder — and therefore one temporal bias — onto all 709 heterogeneous features at once is exactly the design decision the proposed model revisits in Sections 3.4 and 3.5.



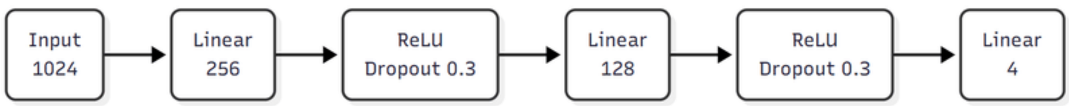
(a) `openface_mlp`: the $T \times 709$ sequence is flattened and classified by an MLP.

(b) `openface_tcn`: a TCN over the 709-d sequence, adaptively average-pooled and classified.

(c) `openface_lstm`: a two-layer LSTM over the 709-d sequence whose final state is classified.



(d) `openface_transformer`: a Transformer encoder over the 709-d sequence, mean-pooled and classified.



(e) `i3d_mlp`: the pooled 1024-d I3D clip vector is classified by an MLP.

Figure 3.2: The five monolithic baseline architectures. Each applies a *single* encoder to the full feature sequence (or pooled I3D vector) before classification.

3.4 Semantic-Group Decomposition

The 709 OpenFace features are partitioned into five behaviourally meaningful groups by matching the column names, in the fixed order used everywhere in the pipeline (gaze, eye, face, head, action units). Table 3.2 summarises the partition; every one of the 709 columns belongs to exactly one group.

Table 3.2: Semantic decomposition of the 709-dimensional OpenFace descriptor.

Group	Content	Dim
Gaze	gaze direction vectors and angles	8
Eye landmarks	2-D/3-D eye-region landmark coordinates	280
Face landmarks	2-D/3-D facial landmark coordinates	340
Head pose	head translation and rotation parameters	46
Action units	facial action-unit presence and intensity	35
Total		**709**

This decomposition is the inductive prior of the method. The five subsystems have different temporal dynamics — rapid, noisy micro-saccades in the gaze and eye groups; slow, smooth drift in head pose; brief, sparse bursts in action units — so each can in principle be matched to an encoder whose temporal bias suits it. Encoding all 709 features together, as the monolithic baselines of Section 3.3 do, forces a single bias on all of them at once.

3.5 Hybrid Architecture (Proposed)

The proposed model, shown in Figure 3.3, encodes each stream independently, fuses the resulting embeddings, and supervises every stream with its own auxiliary head. It is built from the same three encoder families as the OpenFace baselines of Section 3.3 — TCN, Transformer, and LSTM — but where a baseline applies one encoder to the full $T \times 709$ sequence, the hybrid assigns an independently chosen encoder to each of the five semantic groups of Section 3.4, so any difference in performance can be attributed to the decomposition rather than to new encoder machinery.

Per-group encoder. Each group is encoded by a `GroupEncoder` that is one of three interchangeable temporal encoders (Figure 3.4), followed by temporal mean-pooling and a layer normalisation, producing a fixed **64-dimensional** embedding:

- a **TCN** — two residual blocks of dilated causal 1-D convolutions (channel widths $64, 64$; kernel size 3; dropout 0.2);
- a **Transformer** encoder — a linear projection to 64 dimensions, a sinusoidal positional encoding, and two encoder layers (4 heads, feed-forward width 128, GELU, dropout 0.2);
- an **LSTM** — a two-layer recurrent encoder of hidden width 64 (dropout 0.2).

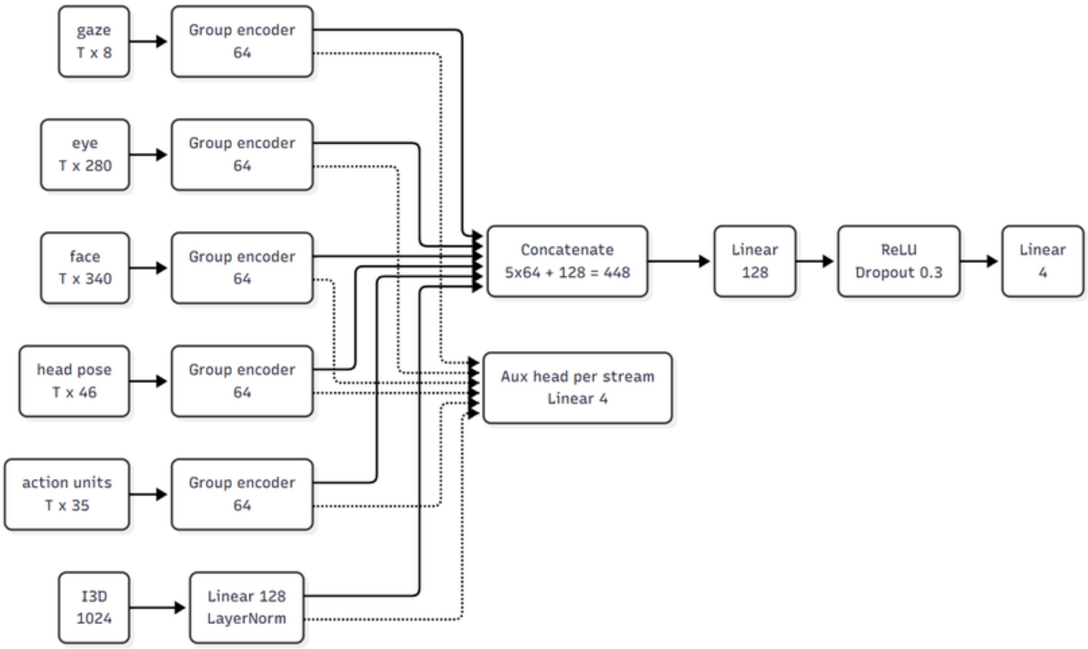
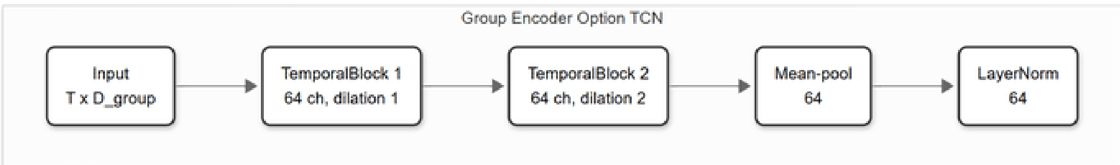
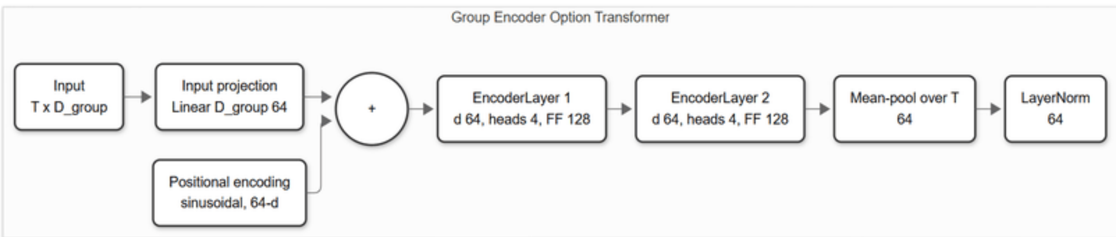


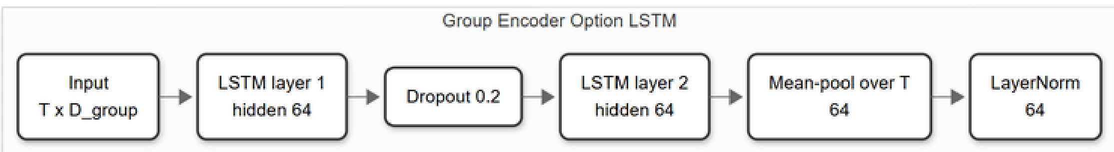
Figure 3.3: Semantic-group hybrid architecture. Each OpenFace group is encoded independently into a 64-d embedding (group encoders are chosen per group from TCN / Transformer (T) / LSTM); the I3D clip vector is encoded by an MLP into a 128-d embedding. The six embeddings are concatenated (448-d) and classified by a shared MLP; each stream also feeds an auxiliary head. The OpenFace-only variant drops the I3D row and fuses a 320-d vector.



(a) TCN group encoder.



(b) Transformer group encoder.



(c) LSTM group encoder.

Figure 3.4: The three interchangeable per-group encoders. Each maps a group's $T \times D_{\text{group}}$ sequence to a 64-dimensional embedding through temporal mean-pooling and layer normalisation; one is chosen independently for each of the five groups.

Fusion and heads. The five 64-d group embeddings are concatenated into a 320-d vector and passed through a two-layer MLP classifier ($320 \to 128 \to 4$, ReLU, dropout 0.3) that produces the main logits. Each group additionally feeds an **auxiliary classification head** ($\text{dropout} \to \text{Linear}(64 \to 4)$). The total training loss is the main-head loss plus `aux_weight` $= 0.2$ times the mean of the per-group auxiliary losses (Section 3.6); the auxiliary supervision forces every stream to be individually discriminative.

Multimodal variant. The I3D-fused model adds a sixth stream: the single 1024-d I3D clip vector is encoded by an MLP ($1024 \to 256 \to 128$, ReLU, dropout 0.3) and layer-normalised to a **128-d** embedding with its own auxiliary head. The fusion MLP then takes a **448-d** input ($5 \times 64 + 128$). The I3D stream uses an MLP rather than a temporal encoder because the provided I3D feature is a single pooled clip vector, not a sequence.

Configuration and `arch_key`. A hybrid configuration is fully specified by the encoder chosen for each of the five groups, written as an `arch_key` of five tokens in group order, for example `TCN_T_TCN_LSTM_T` (`T` = Transformer, `TCN` = TCN, `LSTM` = LSTM). With three choices per group there are $3^5 = \mathbf{243}$ configurations, each evaluated with and without the I3D stream (Section 5.2).

3.6 Loss Functions

The five baseline models are each trained under all three losses introduced in Section 2.5 — cross-entropy, weighted cross-entropy, and the ordinal (EMD) loss of Equation (2.5) — so that the effect of the loss can be measured directly. The class weights for the weighted and ordinal variants are inverse-frequency weights computed on the **training split** and normalised to mean 1: for class c with n_c training samples and C non-empty classes,

$$w_c = \frac{1}{n_c} \cdot \frac{C}{\sum_k \frac{1}{n_k}}, \qquad \text{so that} \quad \frac{1}{C} \sum_c w_c = 1. \tag{3.1}$$

The ordinal loss is applied in its class-weighted form as well, multiplying each sample's CDF distance by w_y.

For the hybrid models the objective also includes the auxiliary term. With main logits z and per-stream auxiliary logits $\{a^{(s)}\}$, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{main}}(z, y) \; + \; \lambda_{\text{aux}} \cdot \frac{1}{S} \sum_{s=1}^{S} \mathcal{L}_{\text{CE}}\big(a^{(s)}, y\big), \tag{3.2}$$

where $\mathcal{L}_{\text{main}}$ is the main-head classification loss, $\lambda_{\text{aux}} = 0.2$ is the auxiliary weight, and the auxiliary heads are always trained with plain cross-entropy, over the S streams (five groups, or six with I3D). The formulation admits any of the three losses for the main head; the 243-configuration hybrid ablation reported here fixes $\mathcal{L}_{\text{main}}$ to cross-entropy, so that its results isolate the effect of the architecture from that of the loss.

3.7 Training Protocol

All models are optimised with Adam [16] at a learning rate of 1×10^{-4}. Training uses mini-batches (128 for the baseline runs, 64 for the 243-configuration ablation) and **early stopping on the validation loss** — the CMOSE `unlabel` split, and its analogues for DAiSEE and Combined — with a patience of 10 epochs and an epoch cap (800 for baselines, 400 for the ablation). The checkpoint with the lowest validation loss is restored for testing. Following the convention adopted in this project, **only the training and validation loss are recorded per epoch**; the final classification metrics are evaluated exactly once, on the held-out test split. Runs use a fixed random seed (42) and optional automatic mixed precision on GPU, and every run converges stably and early-stops well before the cap, so the differences reported between models are architectural rather than optimisation artefacts.

3.8 Evaluation Protocol

Cross-dataset matrix. Generalisation is measured with a 3$\times$3 matrix: every model and loss is trained on each of {CMOSE, DAiSEE, Combined} and evaluated on each of {CMOSE-test, DAiSEE-test, Private}. The diagonal cells are the in-domain results; the off-diagonal and the Private column measure transfer to unseen distributions.

Metrics. Each evaluation reports the six metrics defined in Section 2.6 — accuracy, macro-accuracy, MAE, macro-MAE, Cohen's kappa, and QWK — with **QWK, macro-accuracy, and macro-MAE as primary**; the empirical justification for this choice is laid out in the evaluation-protocol section of Chapter 4 (Section 4.2). Row-normalised confusion matrices and per-class F1 scores are used for error analysis.

Aggregation and reproducibility. The complete study comprises roughly 1,500 trained runs (the five baselines $\times$ three losses $\times$ three sources, and the 243 per-group hybrid configurations with and without I3D); all are aggregated programmatically, and every figure and table of Chapters 4 and 5 is produced from the aggregate by a single command.

3.9 Chapter Summary

This chapter specified the methodology end to end: the pipeline; the three training corpora and their imbalance, the test-only private set and its collection, and the leakage-free preprocessing; the five monolithic baseline architectures; the five-group decomposition of the OpenFace descriptor; and the semantic-group hybrid (Figure 3.3) that reuses the baselines' encoder families per group, with its fusion and auxiliary heads, I3D variant, and 243-configuration design space. It also fixed the loss formulations, the training protocol, and the 3×3 evaluation protocol with its primary metrics, which Chapters 4 and 5 apply to the in-domain and generalisation results.

CHAPTER 4. BASELINE MODELS

4.1 Overview

This chapter studies the *monolithic baseline* models — a feature sequence pushed through a single temporal encoder (Section 3.3) — and fixes the methodological frame the rest of the thesis depends on. Section 4.2 fixes the **evaluation protocol**: the metrics, the development corpus, and the development loss; the two arguments that require out-of-domain evidence — that accuracy cannot detect cross-corpus collapse, and that the loss ranking is stable under shift — are stated there as claims and proved in Section 4.5.2. Under the fixed protocol, Section 4.3 ranks the five baselines and sets the bar the proposed architecture must clear; Section 4.4 descends from scores to individual predictions and asks whether models with similar scores give the *same answers* — the analysis that motivates the hybrid design of Chapter 5; and Section 4.5 takes the baselines out of domain with the 3$\times$3 cross-dataset matrix.

How results are aggregated. Two conventions hold in both results chapters: a *winner* is the **maximum** over the tunable axis (the loss, and in Chapter 5 the per-group encoder assignment), a hyperparameter selected on validation data, while a claim about a *family* of configurations is reported as **means and distributions**. Every configuration is trained once with a fixed seed (Section 3.7), so a reported mean is a mean over the design grid — robustness to design choices, not a multi-seed noise estimate.

4.2 Evaluation Protocol

4.2.1 Metrics

Engagement is an *ordinal*, severely *imbalanced* four-class problem: the levels lie on a line, and 69.4% of CMOSE is the single class *Engage* (Table 3.1). Raw accuracy ignores both properties — it rewards majority-class collapse and counts a one-level miss the same as a three-level miss — whereas QWK is chance-corrected and weights every error by the squared distance between predicted and true levels (Section 2.6). On first principles, accuracy is the wrong yardstick for this task; the in-domain data say the same thing twice over.

First, **the leaderboard winner depends on the metric**. Table 4.1 reports which of the fifteen baseline configurations (five architectures $\times$ three losses) each of the six metrics declares best in-domain on CMOSE, and no two camps agree: QWK crowns the **OpenFace TCN under cross-entropy** (0.537), both macro metrics crown the **I3D MLP under the ordinal loss** (macro-accuracy 0.635, macro-MAE 0.458), and accuracy, micro-MAE, and Cohen's kappa all crown the **I3D MLP under cross-entropy** (0.773, 0.244, 0.436). A metric choice is never innocent, so it must be argued, not defaulted.

Table 4.1: The winning base configuration according to each metric (in-domain CMOSE).

Metric	Best model	Loss	Value
QWK	openface_tcn	CE	0.537
Macro-Accuracy	i3d_mlp	Ordinal	0.635
Macro-MAE (lower is better)	i3d_mlp	Ordinal	0.458
Accuracy	i3d_mlp	CE	0.773
MAE (lower is better)	i3d_mlp	CE	0.244
Cohen κ	i3d_mlp	CE	0.436

Second, **the six metrics form two blocks, bridged only by QWK**. Figure 4.1 reports the Kendall rank correlation between every pair of metrics over the fifteen configurations: a *micro* block — accuracy and micro-MAE ($\tau = 0.98$), dominated by the majority class — and a *macro* block — macro-accuracy and macro-MAE ($\tau = 0.89$) — that are nearly orthogonal to each other (cross-block $\tau \approx 0.01$–0.10). **QWK is the only metric with moderate-to-strong agreement with both blocks** ($\tau = 0.49$–0.60): the compromise metric, sensitive to ordinal distance without being enslaved to the majority class.

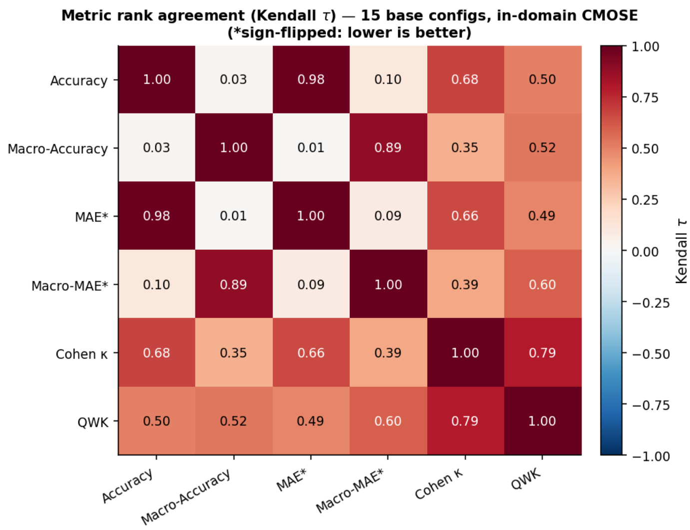


Figure 4.1: Rank agreement (Kendall τ) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive τ means agreement). The metrics form a micro block and a macro block that barely agree; QWK is the only metric correlated with both.

Protocol. The thesis adopts **QWK, macro-accuracy, and macro-MAE** as the three primary metrics, with QWK as the headline; a model is judged strong only when all three agree. Accuracy and micro-MAE remain in the tables for comparability with the literature but never select models. *Deferred evidence:* the decisive argument against accuracy — that it cannot even detect the collapse of a model's ordinal signal under distribution shift — is delivered in Section 4.5.2.

4.2.2 Development Corpus

Table 4.2 compares the two in-domain cells of the evaluation matrix: the best in-domain model reaches **QWK 0.537 on CMOSE but only 0.166 on DAiSEE** (macro-accuracy 0.535 versus 0.289, macro-MAE 0.547 versus 1.226 — more than twice the ordinal error), while DAiSEE's raw accuracy of 0.548 hides the failure — a first taste of the foil accuracy was retained to play. A corpus on which no model rises meaningfully above chance cannot separate good architectures from bad ones: whatever ranking it produces is dominated by label noise. **CMOSE is therefore the development corpus**; DAiSEE re-enters as a transfer source and target in Section 4.5 and in Chapter 5, where its difficulty is informative rather than obstructive.

Table 4.2: Best in-domain result per dataset (CMOSE vs DaiSEE; macro-MAE lower is better).

In-domain dataset	Best model	Loss	QWK	Macro-Acc	Macro-MAE	Accuracy
CMOSE	openface_tcn	CE	0.537	0.535	0.547	0.761
DaiSEE	i3d_mlp	CE	0.166	0.289	1.226	0.548

4.2.3 Development Loss

In-domain, the loss axis is a clean trade within the primary metrics. Figure 4.2 shows, for every baseline, that rebalancing the loss from cross-entropy through weighted cross-entropy to the ordinal loss **raises macro-accuracy and improves macro-MAE** while **QWK falls** for four of the five baselines: the balanced losses buy minority-class recall at the cost of chance-corrected agreement. The trade is genuinely Pareto — no loss is best on all three primaries at once. For the I3D MLP, for instance, switching from cross-entropy to the ordinal loss lifts macro-accuracy from 0.526 to 0.635 and improves macro-MAE from 0.585 to 0.458 — the best balanced figures of any baseline — while QWK falls from 0.519 to 0.487.

Protocol. The strongest ordinal agreement of the whole grid is obtained under **cross-entropy** (best-architecture QWK by loss: CE 0.537, weighted CE 0.500, ordinal 0.487), so **cross-entropy is the development loss**: it maximises the headline metric and isolates the effect of the architecture from the confound of the loss; the

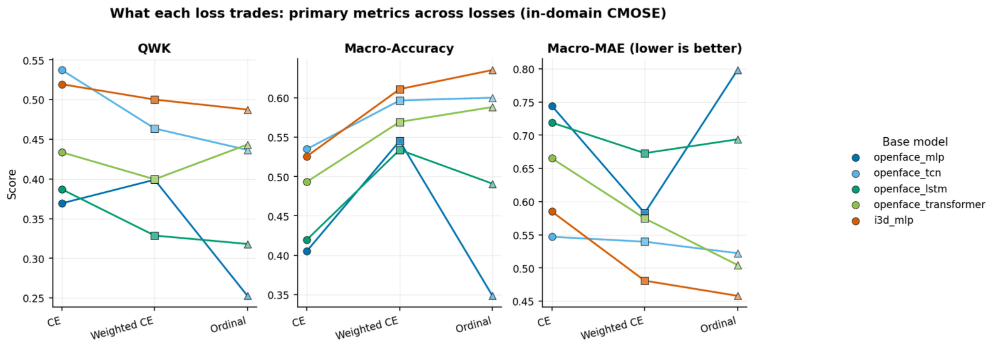


Figure 4.2: What each loss trades: the three primary metrics for every baseline across the three losses, in-domain on CMOSE. Macro-accuracy and macro-MAE improve as the loss is rebalanced, while QWK is generally highest under cross-entropy.

hybrid of Chapter 5 is consequently trained with it. *Deferred evidence:* Section 4.5.2 shows this CE > weighted CE > ordinal ranking is preserved intact under domain shift — which retroactively makes cross-entropy the *reliable* choice, not merely the convenient one. The training and validation loss curves of every configuration converge stably and early-stop well before the epoch cap (Section 3.7), so the differences reported in this chapter are architectural rather than optimisation artefacts.

4.3 In-Domain Results

Five monolithic baselines — four on the OpenFace descriptor (an MLP, a TCN, an LSTM, and a Transformer) and one on the I3D clip vector (the I3D MLP) — are trained and tested in-domain on CMOSE under each of the three losses; Figure 4.3 reports the complete grid on all six metrics. On the primary metrics, the strongest baseline is the **OpenFace TCN under cross-entropy at QWK 0.537** (max-over-loss, per Section 4.1): the bar the proposed architecture of Chapter 5 must clear.

The grid adds two structural observations. First, **the loss axis decides which metric family a model wins**: cross-entropy wins the micro and agreement metrics, while only the rebalanced losses recover the rare disengaged classes that macro averaging weights equally — a direct symptom of the class imbalance. Second, **the architecture axis decides the feature story**: the OpenFace TCN's per-frame facial dynamics carry the cues — a head turning away, gaze drifting off-screen — that separate the rarer disengaged levels and win the ordinal primary, while the I3D MLP, over a single pooled appearance/motion vector, aligns with the majority *Engage* class and tops every micro metric. On the headline metric the two are nearly tied — **QWK 0.537 versus 0.519** — and two near-tied models built on *disjoint feature families* pose a question the score table cannot answer: are they interchangeable, or do they succeed on different clips?

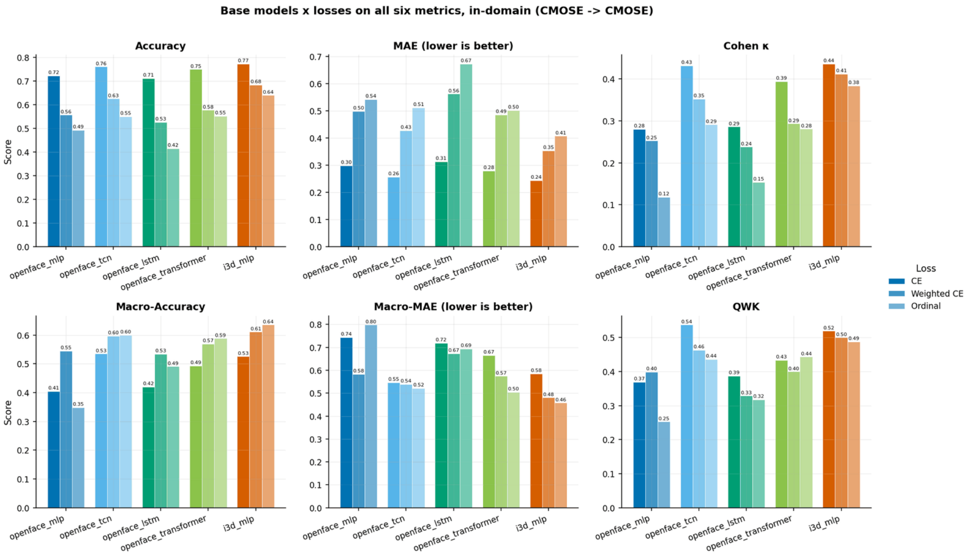


Figure 4.3: The five base models under the three losses on all six metrics, in-domain on CMOSE. The metrics disagree on the winner: accuracy, MAE, and Cohen's kappa favour the I3D MLP under cross-entropy, QWK favours the OpenFace TCN, and the macro metrics peak under the balanced losses.

4.4 Prediction Agreement

Scores summarise *how often* a model is right; they say nothing about *which clips* it is right on. This section descends to individual predictions on the 1,221 in-domain CMOSE test clips, first across the whole grid, then for the two feature families specifically.

4.4.1 Agreement between Configurations

Figure 4.4 reports the pairwise Cohen's kappa between the predictions of every pair of configurations. Mean pairwise agreement is only $\kappa = 0.27$–0.39 — far below what the clustered scores of Figure 4.3 suggest — and sharing the architecture ($\kappa = 0.391$) or the loss ($\kappa = 0.374$) raises it only mildly over sharing neither ($\kappa = 0.273$): the fifteen configurations are genuinely different hypotheses about the data, not reparametrisations of one another.

The practical consequence is **headroom**. The best single configuration classifies 77.3% of the test clips correctly, but **at least one of the fifteen is correct on 97.6%** (the oracle); only 2.4% of clips defeat every model. Yet plain majority voting over the fifteen lands at 73.5% — *below* the best single model — because the errors are too decorrelated for unweighted voting to harvest. The headroom exists, but claiming it requires a *learned* combination of complementary views of the input — precisely the design premise of the semantic-group hybrid of Chapter 5, which fuses separately-encoded behavioural streams instead of averaging finished predictions.

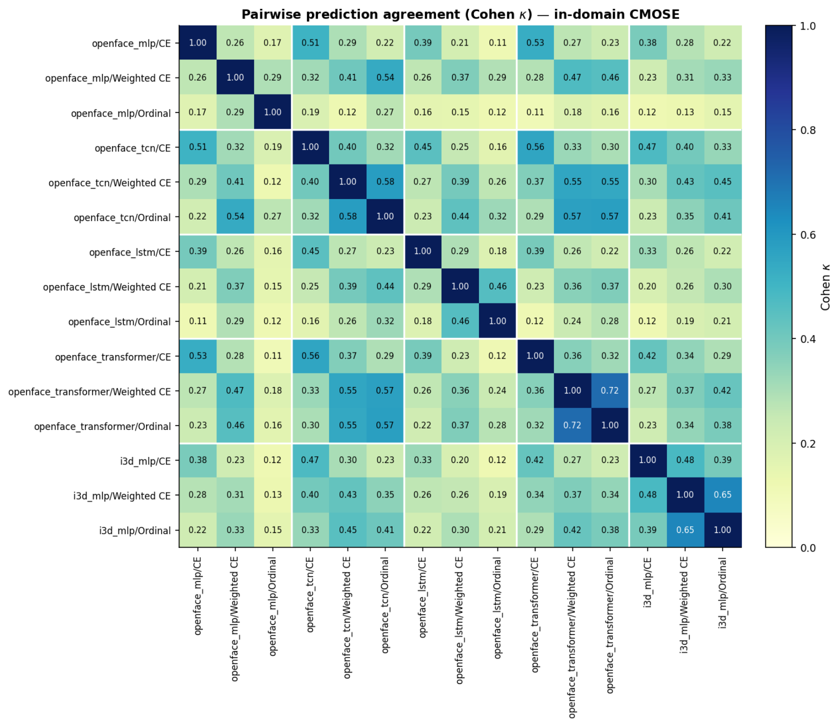


Figure 4.4: Pairwise prediction agreement (Cohen's κ) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks. The diagonal blocks are the same architecture under different losses; agreement is moderate at best everywhere.

4.4.2 OpenFace versus I3D

The two families behind the near-tie of Section 4.3 are of very different natures: **OpenFace** is a per-frame, structured account of facial behaviour (gaze, eye and face landmarks, head pose, action units), while **I3D** is a single clip-level vector of global appearance and motion. Table 4.3 compares their best cross-entropy models: the OpenFace TCN wins all three primary metrics, so by scores alone the I3D MLP looks merely redundant.

Table 4.3: Best OpenFace encoder vs the I3D MLP on the primary metrics (in-domain CMOSE, cross-entropy; macro-MAE lower is better).

Feature family	Model	QWK	Macro-Acc	Macro-MAE
OpenFace (behaviour descriptors)	openface_tcn	0.537	0.535	0.547
I3D (appearance/motion)	i3d_mlp	0.519	0.526	0.585

The agreement map says otherwise. In Figure 4.4, the I3D MLP's row is the palest in the grid: its agreement with the OpenFace TCN is among the *lowest* of any pair, despite the two posting the closest scores. Two models that score alike but agree this rarely must succeed on different clips, and the clip-level decomposition confirms it: against the OpenFace TCN, the I3D MLP has the largest exclusivecorrect share of any pair in the grid — **8.6% of all test clips are clips only I3D gets right** — and the smallest both-wrong share (15.3%). The pair's oracle accuracy is 84.7%, +7.4 points over the best single model and the largest pair gain in the grid.

Conclusion. The behaviour descriptors and the appearance features are looking at genuinely **different evidence**: the families fail on different clips. This is the empirical motivation for *fusing* the two streams in the hybrid of Chapter 5 — a motivation the score table alone could never supply.

4.5 Cross-Dataset Generalization

In-domain numbers are not the deployment question. This section trains on one corpus and tests on another, with three aims: measure whether any ordinal signal survives a change of corpus, close the two arguments the protocol deferred, and ask whether in-domain strength predicts generalisation at all.

4.5.1 The 3×3 Matrix

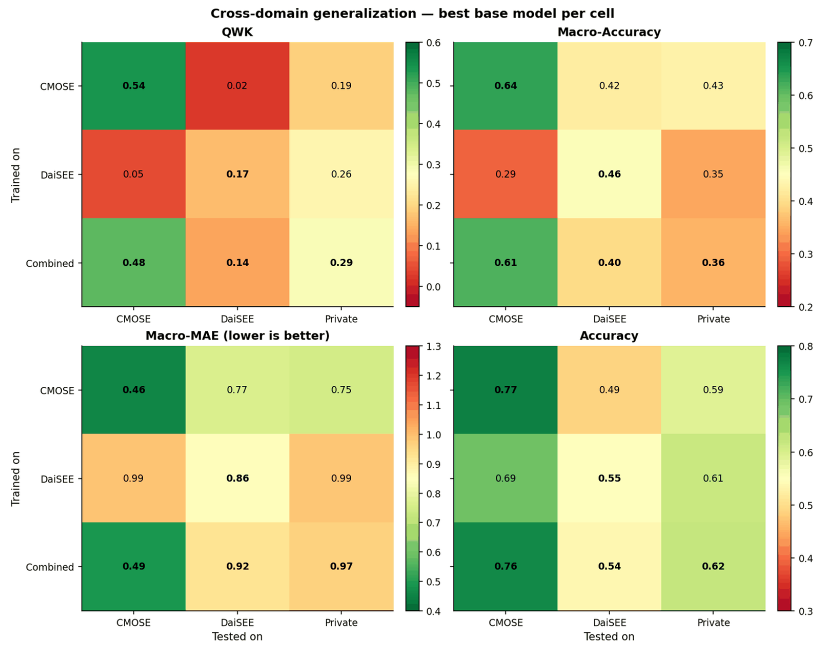


Figure 4.5: Cross-domain generalisation, best base model per cell, on the three primary metrics — QWK, macro-accuracy, and macro-MAE (lower is better) — plus raw accuracy, the deceptive foil: accuracy stays high in exactly the off-diagonal cells where every chance-corrected metric collapses.

The collapse. The 3×3 heatmaps of Figure 4.5 are the central baseline generalisation result. **On the diagonal** (train and test on the same corpus) QWK is healthy (CMOSE 0.54), macro-accuracy is well above chance (CMOSE 0.64), and ordinal error is low (CMOSE macro-MAE 0.46). **Off the diagonal all three collapse together**: CMOSE→DAiSEE QWK $= 0.02$ and DAiSEE→CMOSE $= 0.05$ (chance-level agreement), macro-accuracy falls toward the 0.25 chance floor, and macro-MAE inflates toward a full class. The collapse is roughly symmetric — neither public corpus transfers to the other — which rules out one corpus being a superset of the other and points to a genuine shift in both label priors and recording conditions.

The architecture ranking is reshuffled. Comparing, per architecture, the in-domain QWK with the mean QWK over the *unseen-target* cells — the cells whose test corpus contributed no training data (the two cross-corpus cells and the private column) — the in-domain order TCN 0.537 > I3D 0.519 > Transformer 0.443 > MLP 0.399 > LSTM 0.387 becomes TCN ≈ Transformer > LSTM > **I3D** > MLP on unseen targets. **The I3D MLP falls from second place to fourth**: its global appearance features bind to the corpus, while the OpenFace behaviour descriptors carry more corpus-invariant signal.

The lever: pool the source corpora. The most actionable cell is the unseen **Private** column: training on the **Combined** corpus gives the best private-set QWK (0.285), beating DAiSEE-only (0.256) and CMOSE-only (0.190), while keeping near-in-domain CMOSE performance (0.477, the Combined→CMOSE cell of Figure 4.5). The union of CMOSE and DAiSEE spans a wider range of label priors than either alone, so an unseen target sits closer to the pooled distribution. This is the single most effective design lever for an unknown target, and Chapter 5 combines it with the architecture contribution on the decisive private set (**RQ4**).

4.5.2 Closing the Loop: Metric and Loss Reliability

Accuracy cannot see the collapse (completes Section 4.2.1). In the accuracy panel of Figure 4.5, CMOSE→DAiSEE posts a deceptively reasonable 0.49 and DAiSEE→CMOSE 0.69 — in the very cells where the QWK panel reads 0.02 and 0.05, because predicting near the majority class scores well on any imbalanced target. A metric that reports business-as-usual while the ordinal signal is gone is a blindfold, not a safety net: **accuracy is the least reliable metric in this study and QWK** — the only block-bridging metric, and the one that detects the collapse — **the most**.

The loss ranking survives the shift (completes Section 4.2.3). On the unseen-target cells the best-architecture QWK by loss follows the same CE > weighted CE > ordinal order as in-domain. The loss is the only factor in the grid whose ranking transfers (the architecture ranking reshuffles): choosing cross-entropy in-domain remains the right choice out of domain, which retroactively validates fixing it as the development loss.

4.5.3 In-Domain versus Transfer

Does in-domain strength predict transfer? Across the thirty trained baseline configurations, the Spearman rank correlation between in-domain QWK and mean QWK over the unseen-target cells is only $\rho = 0.28$ — too weak to select models on, so model selection for an unknown target must lean on the pooling lever rather than the in-domain leaderboard. Chapter 5 sharpens this warning on the hybrid population, where the correlation even turns negative (Figure 5.6).

4.6 Chapter Summary

The evaluation protocol fixed **QWK, macro-accuracy, and macro-MAE** as the primary metrics, **CMOSE** as the development corpus, and **cross-entropy** as the development loss; under it, the strongest baseline is the OpenFace TCN at **QWK 0.537** — the bar the hybrid must beat. The prediction-level analysis showed the configurations agree far less than their scores suggest (oracle 97.6%) and that the OpenFace and I3D families fail on *different* clips, the direct motivation for fusing them; the cross-dataset study showed baselines do not transfer off-the-shelf, vindicated the metric and loss choices, and made corpus pooling the most effective simple mitigation. Chapter 5 now asks whether learning to fuse the complementary streams beats these baselines, and whether the gain compounds with pooling on unseen data.

CHAPTER 5. SEMANTIC-GROUP HYBRID

5.1 Overview

Chapter 4 fixed the evaluation frame — QWK, macro-accuracy, and macro-MAE as primary metrics; CMOSE as development corpus; cross-entropy as development loss; the OpenFace TCN at QWK 0.537 as the bar — and supplied the design motivation: the baseline configurations, and the OpenFace and I3D feature families in particular, fail on different clips. This chapter evaluates the proposed **semantic-group hybrid** (Section 3.5) against that frame: all $3^5 = 243$ per-group encoder configurations, each with and without the I3D stream, trained with cross-entropy. Section 5.2 first re-validates the Chapter 4 frame on this 486-configuration population (RQ1 in part); Section 5.3 isolates which behavioural group's encoder choice matters (RQ2); Section 5.4 measures the effect of I3D fusion with a paired ablation (RQ3); Section 5.5 pins down the headline in-domain comparison (RQ1); and Section 5.6 takes the hybrid out of domain, establishing that it generalizes better than the baselines — in *every* cell of the matrix, and most of all on the decisive self-collected private set, where the architecture and the corpus-pooling lever *compound* (RQ4). Section 5.7 discusses why. As in Chapter 4, family-level claims are reported as means and distributions and headline comparisons as the tuned maximum (Section 4.1).

5.2 Consistency with the Baseline Frame

The ablation population doubles as a stress test of Chapter 4's protocol: 486 configurations of a different architecture family, none of which existed when the frame was fixed. Figure 5.1 plots the distribution of the two hybrid families (with and without I3D) on every metric, each panel against the best base model on that metric. Three findings of Chapter 4 reproduce.

The metric structure reproduces. Recomputing the Kendall rank correlation between the six metrics over the 486 in-domain configurations yields the same two-block picture as Section 4.2.1: a micro block, a macro block, weak cross-block agreement, and QWK again the only bridge between them — a property of the metrics on this task, not of any model family.

Accuracy is still blind. In Figure 5.1 every configuration sits in a narrow accuracy band around the best baseline's 0.773 (top-left panel), so a model chosen on accuracy would see no architecture signal at all, while the QWK panel separates the families cleanly. (The macro panels' dashed lines belong to the *ordinal-loss* I3D MLP, a balanced-loss operating point that buys those figures by giving up QWK — the loss trade-off of Section 4.2.3, not an architecture effect.)

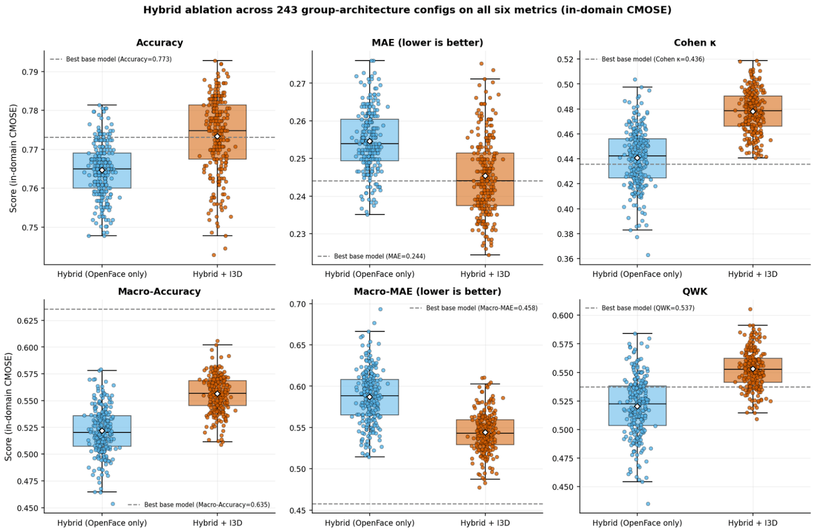


Figure 5.1: Hybrid ablation across all 243 per-group encoder configurations, with and without I3D, on all six metrics (in-domain CMOSE). Each dashed line is the best base model on that metric, across all three losses.

The cross-domain collapse reproduces. Taken out of domain (Section 5.6.1), the hybrid matrix shows the same shape as the baseline matrix: a healthy diagonal (0.605 on CMOSE, 0.228 on DAiSEE) and an off-diagonal collapse, with accuracy again masking it. The collapse is a property of the *problem* — divergent label priors and recording conditions — not of a particular architecture.

All three pillars of the Chapter 4 frame thus survive a 32-fold larger model population; the protocol carries over without amendment (RQ1's frame condition).

5.3 Encoder Choice per Group

The 243-configuration grid varies five group→encoder choices at once; this section marginalises them to find which group's choice actually moves the primary metrics (RQ2), and whether any preference is corpus-specific or survives shift.

Table 5.1 reports the marginal in both regimes. **In-domain**, only **head pose** shows a clear preference: TCN (mean QWK 0.545), ahead of LSTM (0.541) and Transformer (0.523), a spread of 0.022; macro-MAE agrees, lowest (best) under the TCN. The other four groups are essentially encoder-agnostic (QWK spread ≤ 0.006). **Under shift** (the same marginal over the unseen-target cells), head pose again has the largest spread (0.010) and again prefers the TCN, while the other groups' spreads (0.002–0.004) flip winners arbitrarily — noise.

The consistency across regimes elevates the head-pose rule from a tuning accident to a property of the signal: head dynamics such as nodding and turning away are temporally local and convolution-friendly (RQ2). The design space is otherwise flat, so most of the architecture's benefit must come from *decomposing and fusing*

Table 5.1: Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells.

Group	Regime	TCN	Transformer	LSTM	Spread	Prefers
Gaze	In-domain	0.539	0.537	0.534	0.006	TCN
Gaze	Unseen targets	0.093	0.095	0.094	0.002	Transformer
Eye landmarks	In-domain	0.538	0.537	0.535	0.003	TCN
Eye landmarks	Unseen targets	0.096	0.092	0.094	0.004	TCN
Face landmarks	In-domain	0.537	0.534	0.539	0.006	LSTM
Face landmarks	Unseen targets	0.092	0.096	0.093	0.004	Transformer
Head pose	In-domain	0.545	0.523	0.541	0.022	TCN
Head pose	Unseen targets	0.100	0.091	0.090	0.010	TCN
Action units	In-domain	0.536	0.537	0.537	0.001	Transformer
Action units	Unseen targets	0.093	0.096	0.093	0.003	Transformer

rather than from the per-group encoder search — which is what the next section measures.

5.4 Effect of I3D Fusion

Chapter 4 predicted, from the error-overlap analysis of Section 4.4.2, that fusing the I3D stream should pay. This section measures the prediction with the strongest design in the grid: **pairing each OpenFace-only configuration with its exact +I3D twin** (same five encoders), so the distribution of paired QWK differences *is* the I3D effect, free of the configuration as a confounder, in three evaluation regimes.

Where the test corpus is seen in training, fusion pays decisively. Over the seen-target cells the mean paired gain is **+0.060 QWK, and 94% of the 972 pairs improve** (Table 5.2). In-domain on CMOSE this is the cash value of the complementarity of Section 4.4.2: the I3D-fused family is centred above the OpenFace-only family (median QWK 0.553 against 0.522, visible in Figure 5.1), and **82% of the 243 fused configurations clear the 0.537 baseline bar**, against 26% of the OpenFace-only ones. The gain is a property of the *family*, not of one lucky configuration (RQ3).

Table 5.2: Paired effect of adding the I3D stream on QWK, by evaluation regime (each OpenFace-only hybrid against its exact +I3D twin).

Regime	Pairs (n)	Mean ΔQWK	Pairs improving (%)
Seen target	972	+0.060	94
Cross-corpus (public)	486	-0.021	26
Private (unseen)	729	+0.005	52

Beyond the training corpora, the gain evaporates. Cross-corpus the mean paired effect is -0.021 **QWK with only 26% of pairs improving**; on the private set it is +0.005 with 52% improving and high variance across configurations (Table 5.2). I3D's global appearance features drag corpus bias along — the same mechanism behind the I3D MLP's fall from second to fourth in the architecture ranking under shift (Section 4.5.1). The practical rule (RQ3): *fuse I3D when the target domain is represented in training; rely on the OpenFace groups otherwise* — a rule about the family average (the single best private-set model does carry I3D, but it is one tail of a wide, zero-centred distribution).

5.5 Comparison with Baselines

Best hybrid versus best baseline. Table 5.3 lists the top-5 configurations; the best overall is the I3D-fused hybrid `TCN_T_TCN_LSTM_T`, and every top configuration mixes at least two encoder types. On all three primary metrics the best hybrid improves on the best baseline — **+0.068 QWK** (0.605 versus 0.537), macro-accuracy 0.582 versus 0.535, and macro-MAE 0.489 versus 0.547 (lower is better). All three move in the same direction, so the gain is real — but **modest**; it is the CMOSE diagonal cell of the cell-by-cell comparison in Section 5.6.1 (Figure 5.4).

Table 5.3: Top-5 hybrid configs in-domain (CMOSE), sorted by QWK (macro-MAE lower is better).

Variant	Arch (gaze_eye_face_head_au)	QWK	Macro-Acc	Macro-MAE	Accuracy
Hybrid + I3D	TCN_T_TCN_LSTM_T	0.605	0.582	0.489	0.784
Hybrid + I3D	T_TCN_TCN_LSTM_T	0.591	0.606	0.483	0.787
Hybrid + I3D	T_T_LSTM_LSTM_TCN	0.591	0.563	0.505	0.781
Hybrid + I3D	T_TCN_T_TCN_T	0.588	0.566	0.531	0.789
Hybrid + I3D	LSTM_LSTM_TCN_LSTM_T	0.588	0.567	0.517	0.787

Where does the gain land? The per-class F1 scores (Figure 5.2) localise it. Relative to the best baseline, the hybrid improves recall on the ordinal extreme **Highly-Engage** ($0.38 \rightarrow 0.51$) and on **Disengage** ($0.45 \rightarrow 0.55$), while keeping *Engage* near 0.90. The rarest class **Highly-Disengage**, however, stays unsolved and is slightly traded away (recall $0.40 \rightarrow 0.37$): the model reallocates capacity toward the more frequent extreme.

The in-domain improvement is therefore real but **uneven**: the hybrid clears the baseline on all three primary metrics by strengthening the well-populated upper-engagement region and the middle disengaged class, while the $\sim$2–3%-frequency *Highly-Disengage* class remains the principal open challenge (RQ1). The next section shows the picture becomes more severe still out of domain — and that

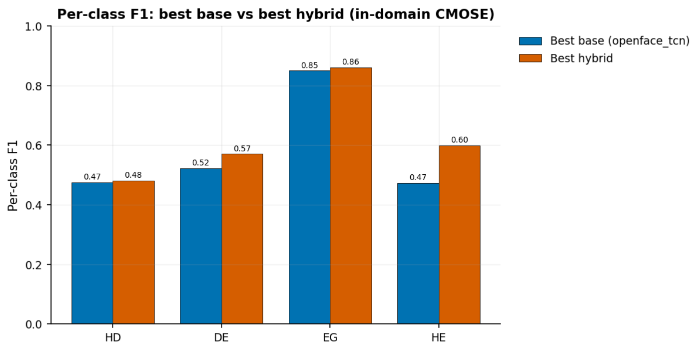


Figure 5.2: Per-class F1: best base versus best hybrid (in-domain CMOSE).

the architecture's advantage over the baseline *widens* there.

5.6 Cross-Dataset Generalization

5.6.1 Cross-Dataset Matrix

The hybrid cross-dataset heatmap (Figure 5.3) shows the same pattern as the baseline matrix of Section 4.5: a healthy diagonal and an off-diagonal collapse of all three primary metrics — the hybrid does not rescue cross-corpus transfer. **The accuracy deception persists**: on DAiSEE→CMOSE the best QWK any of the 486 hybrid runs achieves is 0.10, yet hybrids in that cell post accuracies up to 0.69. A second architecture family thus lands in the same place — the metric frame is not an artefact of the monolithic models — and this is the last time accuracy appears in the thesis's analysis.

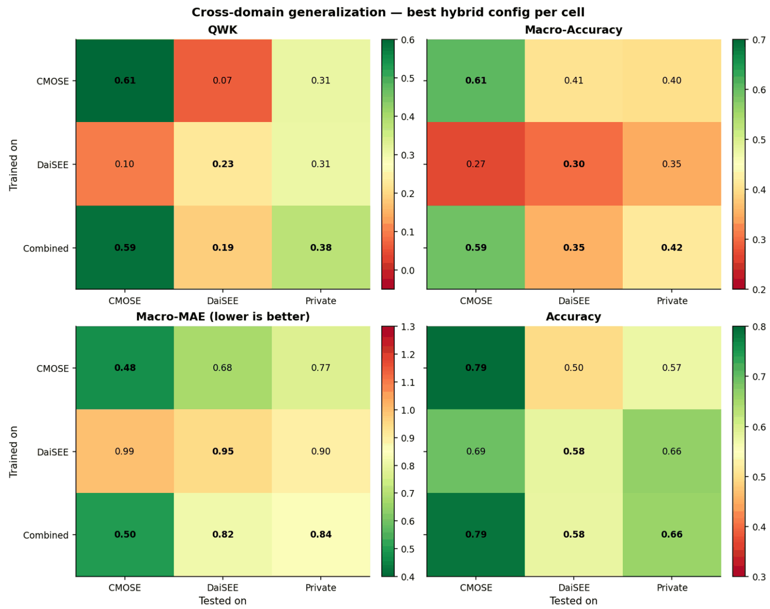


Figure 5.3: Cross-domain generalisation, best hybrid per cell, on the three primary metrics — QWK, macro-accuracy, and macro-MAE (lower is better) — plus raw accuracy, which again stays high in the off-diagonal cells where the chance-corrected metrics collapse.

Within the collapse, the hybrid wins every cell. Figure 5.4 subtracts the baseline matrix (Figure 4.5) from the hybrid matrix (Figure 5.3) cell by cell: the best hybrid beats the best baseline in **all nine train$\times$test cells**, with every cell of the ΔQWK matrix positive. The gains are smallest on the in-domain diagonal ($+0.068$ on CMOSE) and **largest on the decisive Private column** ($+0.118$, $+0.050$, $+0.094$ from the three training sources). The improvement is not a single tuned point but a uniform shift of the entire matrix — the strongest form the family-level claim can take — and, read column by column, the figure makes the chapter's two headline comparisons one object: its CMOSE diagonal cell is the in-domain gain of Section 5.5, and its Private column is the private-set result developed next.

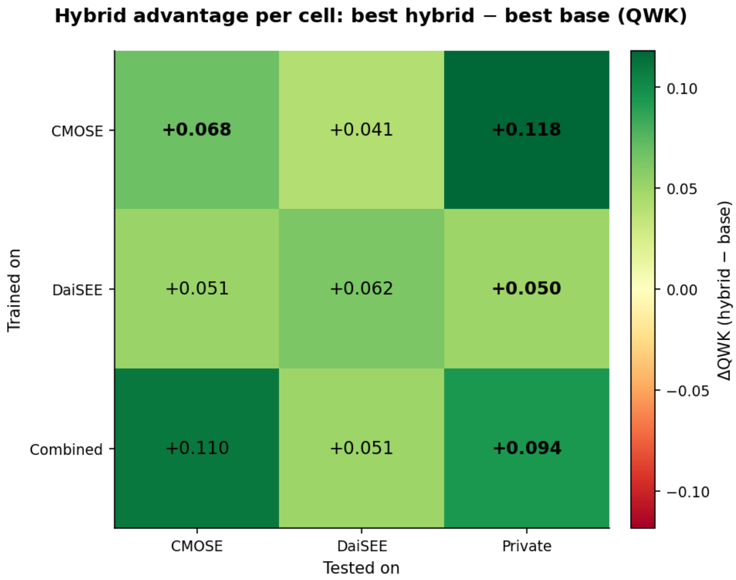


Figure 5.4: Cell-by-cell hybrid advantage: best-hybrid minus best-base QWK over the 3×3 train×test matrix (Figure 5.3 minus Figure 4.5). Every cell is positive — the hybrid wins the whole matrix — and the gain is largest on the unseen, self-collected Private column. The in-domain CMOSE cell and the Private column (bold) are the two comparisons read in Sections 5.5 and 5.6.2.

5.6.2 The Private Set

The most consequential evaluation in this thesis is on the **private set** — the 366 clips we collected and hand-labelled ourselves (Section 3.2.2), processed through the identical OpenFace/I3D pipeline and used **for testing only**. Because no model is ever trained on it, it is a genuine "in-the-wild" probe of deployment, with its own recording conditions and label prior (58% Engage, 31% Highly-Engage, only 2.7% Highly-Disengage); the *only* design lever is the choice of training source. The Private column of Figure 5.4 and Table 5.4 report the best base model and the best hybrid per training source, and two effects compound.

Table 5.4: Private set (test-only): best base vs best hybrid by training source (macro-MAE lower is better).

Train source	Best base (model/loss)	Base QWK	Base macro-MAE	Best hybrid (arch)	Hybrid QWK	Hybrid macro-acc	Hybrid macro-MAE
CMOSE	openface_transformer/ce	0.190	0.750	Hybrid (OpenFace only) TCN_T_LSTM_TCN_T	0.309	0.385	0.914
DaiSEE	i3d_mlp/ce	0.256	1.031	Hybrid + I3D LSTM_LSTM_TCN_LSTM_LSTM	0.306	0.350	0.966
Combined	openface_transformer/ce	0.285	0.997	Hybrid + I3D T_T_T_LSTM_T	0.379	0.385	0.877

1. **Training source matters, and the combined corpus wins.** For both model families, training on **Combined** generalises best to the private set (best base QWK: Combined 0.285 versus DAiSEE 0.256 versus CMOSE 0.190) — pooling heterogeneous source corpora is the most effective simple route to an unseen target, exactly as Chapter 4 predicted.

2. **The hybrid generalizes better — and helps more here than in-domain.**

The hybrid beats the best base model on QWK for *every* training source, and the gap is largest exactly where it matters: under combined training, **QWK 0.379 versus 0.285**, a $+0.094$ improvement — wider than the $+0.068$ in-domain gain, so distribution shift *amplifies* rather than erodes the architecture's advantage. The overall best private-set model is the combined-trained I3D-fused hybrid `T_T_T_LSTM_T`, best on both ordinal primaries: **QWK 0.379** and the lowest **macro-MAE 0.877** (against the best base's 0.997). (One caveat in Table 5.4: the CMOSE-only base model posts a deceptively low macro-MAE (0.75) despite the weakest QWK by predicting conservatively around the majority classes — the false comfort that reporting all three primaries together is designed to catch.)

Figure 5.5 compares the confusion of the two combined-trained models on the private set. The hybrid captures the ordinal structure markedly better: *Engage* recall rises from 0.46 to 0.72 and errors concentrate on the diagonal and its neighbours, where the base model scatters *Engage* mass into *Highly-Engage* (0.44). As in-domain, the rare disengaged classes are the failure mode — hybrid *Disengage* recall is only 0.29, and **Highly-Disengage collapses to 0.00** (all ten private HD clips are misread).

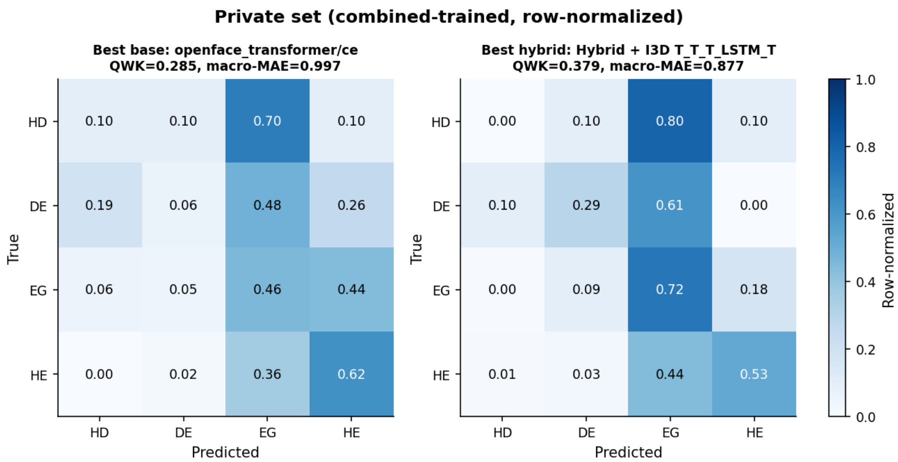


Figure 5.5: Private-set confusion of the combined-trained best base model versus best hybrid (row-normalised).

5.6.3 In-Domain versus Transfer

Section 4.5.3 showed that a baseline's in-domain QWK barely predicts its transfer ($\rho = 0.28$); the ablation population sharpens the warning. Figure 5.6 overlays the 972 trained hybrid models on the baseline scatter: within the hybrid family the correlation is **negative** ($\rho = -0.31$) — among closely related configurations, squeezing out more in-domain QWK mildly *hurts* transfer, because the extra fit is corpus fit. The hybrid's advantage is therefore not a position on an in-domain–

transfer trend — no usable trend exists — it is a **lift of the whole trade-off**: the hybrid cloud sits above the baselines at every level of in-domain performance, and the private-set winner is found by family-level robustness and pooled training, not by picking the best in-domain configuration.

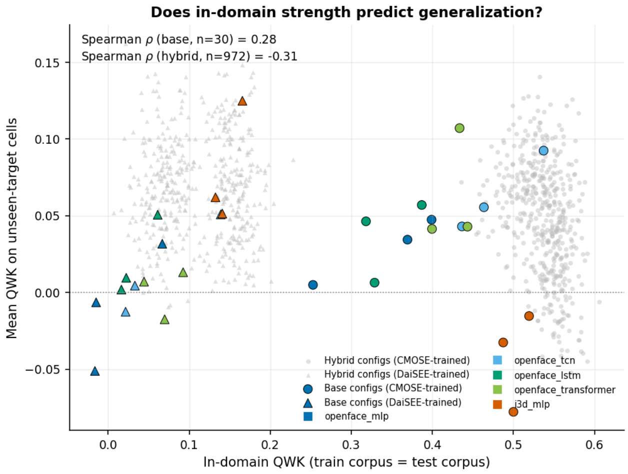


Figure 5.6: In-domain QWK versus mean unseen-target QWK with the hybrid ablation population overlaid on the baseline configurations. Within the hybrid family the correlation is negative ($\rho = -0.31$): more in-domain fit means more corpus fit. The hybrid cloud sits above the baselines throughout — the architecture lifts the trade-off rather than riding it.

5.7 Discussion

Why the levers work. Pooling works through prior coverage: CMOSE and DAiSEE are imbalanced in opposite directions (Section 3.2), so their union spans a wider range of label priors and an unseen target sits closer to the pooled corpus than to either single source. The hybrid's margin *widens* off-domain for the reason it was designed: encoding each behavioural channel separately, each forced to be discriminative through its auxiliary head, yields a representation less tied to any one corpus's idiosyncrasies than a monolithic encoder free to latch onto a corpus-specific shortcut — the negative in-domain–transfer correlation ($\rho = -0.31$) is the same mechanism from the other side. The two feature families diverge under shift because the OpenFace descriptors live in subject-centred physical units that mean the same thing in any setup, whereas the I3D embedding encodes global appearance — exactly what changes between corpora — so fusion adds signal only where the test appearance statistics are represented in training.

What the headline number means. QWK 0.379 is a clear improvement over chance, every single-source model, and the best baseline ($+0.094$), but in absolute terms *moderate*: with macro-MAE 0.877 the model captures the broad engaged-versus-disengaged trend yet is unreliable at fine distinctions, and the rarest class *Highly-Disengage* fails in both regimes. It is usable as an aggregate, classroom-level signal but not to flag an individual disengaged learner without human confirmation; the open problem is one of data and objective — no encoder assignment can recover a class the training distribution barely contains — which is why the imbalance-specific remedies of Chapter 6 are the natural next step.

5.8 Chapter Summary

The Chapter 4 frame survived the 486-configuration stress test: the metric block structure, the blindness of accuracy, and the cross-domain collapse all reproduced as task properties. Only head pose has a clear encoder preference (a TCN), so the benefit comes from decomposing and fusing rather than the encoder search (RQ2); I3D fusion is decisively positive where the test corpus is seen in training, neutral-to-harmful beyond it (RQ3). In-domain the best hybrid beats the best baseline on all three primary metrics (QWK 0.605 versus 0.537, RQ1); out of domain it wins *every* cell of the matrix, compounding with combined-corpus training to the best private-set model (QWK 0.379, $+0.094$ over the baseline — wider than in-domain), and lifts the whole in-domain–transfer trade-off rather than riding it (RQ4), with Highly-Disengage the principal failure mode in both regimes. Chapter 6 draws the findings together, revisits the research questions, and outlines future work.

6.1 Summary

This thesis studied four-class ordinal student-engagement recognition from per-frame OpenFace facial-behaviour features and clip-level I3D motion features. It set out to improve recognition over standard monolithic models and, just as much, to describe the problem honestly — in particular its strong, uneven class imbalance and its behaviour under distribution shift.

For the method, the thesis built a complete and reproducible pipeline: feature extraction and leak-free preprocessing; a **semantic-group hybrid temporal architecture** that splits the 709-dimensional OpenFace features into five behavioural groups (gaze, eye landmarks, face landmarks, head pose, action units), encodes each group with its own temporal encoder, and optionally adds a global I3D stream while training every stream with a small head; five monolithic baselines; three loss functions; and a 3×3 cross-dataset evaluation protocol. The central idea was tested by training and evaluating all $3^5 = 243$ per-group encoder choices, with and without I3D, and by a **self-collected, test-only private set** of 366 hand-labelled clips that serves as the deciding deployment test.

The evidence supports five conclusions.

1. **The I3D-fused hybrid is the best in-domain model and, as a family, beats the strongest monolithic baseline.** The I3D-fused family is centred above the baseline QWK of 0.537 (median 0.553, with 82% of its 243 configurations clearing the baseline), and the best configuration reaches **QWK 0.605** with a mixed TCN/Transformer/LSTM assignment (`TCN_T_TCN_LSTM_T`). The gain over the best baseline is real but **small** ($+0.068$ QWK); the value lies in the framework and the full evidence, not in a jump in state of the art.

2. **Most groups don't care which encoder they get; only head pose shows a clear preference, for a TCN** (a 0.022 QWK spread across the three encoders, against ≤ 0.006 for the other four groups). This is an easy-to-read design rule that reflects the short, local nature of head motion.

3. **Engagement models do not generalise across datasets.** Off-diagonal QWK drops to near zero (CMOSE→DAiSEE 0.02) even when accuracy looks fine, so ordinal agreement metrics — not accuracy — must be reported on imbalanced engagement data.

4. **On the self-collected, hand-labelled private set — the most realistic test**

— the three contributions compound. The hybrid's advantage is not limited to one comparison: it beats the best baseline in *all nine* train$\times$test cells of the cross-dataset matrix. Because the private set is test-only, the only choice left is the training data; training on **both datasets together** with the **semantic-group hybrid** gives the best result (QWK 0.379), beating the best baseline by $+0.094$ (against $+0.068$ in-domain) and topping every single-dataset hybrid. This is the practical payoff of the thesis.

5. **The rare Highly-Disengage class is still the main open problem.** It is around 2–3% of the data, is traded away rather than solved by the hybrid in-domain, and drops to zero recall under distribution shift.

A note on scope: the architecture ablation is based on CMOSE because DAiSEE's in-domain ordinal signal is too weak (QWK 0.166) to tell architecture differences apart from label noise; CMOSE is therefore used for development and the private set for the deciding check.

6.2 Synthesis

Read together, the two results chapters tell a single story whose lesson is not the one a benchmark paper usually reports. The thesis has a *building* side — the semantic-group hybrid and the dataset-pooling recipe — and a *critical* side — an honest account of imbalance, non-transfer, and the metrics that show them — and the most important finding lies in how the two work together.

The architecture earns its value off-domain, not on it. In-domain the semantic-group decomposition is a small, easy-to-read improvement: the I3D-fused family sits above the baseline but the best configuration beats it by only $+0.068$ QWK, and most of the 243-cell design space doesn't care which encoder it gets. On its own, that is a small result. What makes it worth building is that the same decomposition pays off *more* exactly where it is hardest: on the unseen private set the gap over the baseline grows to $+0.094$, and it compounds with pooling rather than fighting it. So the contribution is best understood not as a new in-domain state of the art but as a representation that breaks down more slowly under distribution shift — which, for a problem whose only real test is deployment, is the property that matters.

The fusion was measured before it was built. The choice to combine the OpenFace and I3D families did not rest on a hunch: the clip-level agreement analysis of Chapter 4 showed that the best OpenFace model and the I3D baseline disagree more with each other than OpenFace models do among themselves, and that each family gets clips right that the other cannot — evidence of different errors that the paired I3D ablation of Chapter 5 then confirmed in-domain. The same analysis also measured unused room to improve: a perfect picker over the thirty baseline configurations would reach 97.6% of clips against 77.3% for the best single model, which both motivates fusion and marks ensemble selection as a direction the thesis left aside on purpose.

The flat design space is itself the message. That four of five behavioural groups don't care about their encoder, while only head pose prefers a TCN, bounds the contribution honestly: the gain comes from *splitting and fusing* the features, not from a clever per-group encoder search. This is a more transferable design rule than a single tuned configuration, and it is why the family-level effect, rather than any one cell, is reported as the result.

The negative results are contributions, not side notes. The near-total collapse of cross-dataset transfer, the way plain accuracy hides it, and the steady failure on the rarest disengaged class are not minor limits of these particular models; they are properties of the problem that the field's in-domain, accuracy-led habit leaves unmeasured. Showing them with numbers — and showing that QWK, macro-accuracy, and macro-MAE make them visible — changes engagement recognition from "beat the benchmark" to "report ordinal agreement and watch rare-class failure," which is the practice the rest of this chapter recommends.

6.3 Revisiting the Research Questions

The four research questions posed in Section 1.3 can now be answered directly.

RQ1 (decomposition). *Yes, decomposition helps.* Encoding the five behavioural groups separately and fusing them beats encoding all 709 features with a single monolithic encoder: as a family the hybrid sits above the best baseline, and the effect holds across many configurations rather than coming from a single lucky one, with most of the I3D-fused configurations beating the baseline.

RQ2 (encoder assignment). *Only head pose matters.* Four of the five groups basically don't care whether their encoder is a TCN, a Transformer, or an LSTM (spread ≤ 0.006 QWK); head pose alone shows a clear, easy-to-read preference for a TCN. The practical point is that the design space is mostly flat and that a TCN is a safe default for every group, with head pose the one place the choice is worth making on purpose.

RQ3 (motion fusion). *Yes in-domain, with an honest warning under shift.* When the model is tested on data like its training data, adding the clip-level I3D embedding clearly helps: the paired comparison over the seen-target cells shows a mean gain of $+0.060$ QWK with 94% of pairs improving, and the fused family gives both the best in-domain model and the best private-set model. Under cross-dataset shift, however, the same stream is neutral or slightly harmful on average (mean -0.021 on cross-public cells, $+0.005$ on the private set) — the I3D stream carries useful but dataset-specific look cues. So the fusion is recommended, but its gain should be claimed only for the setting where it holds.

RQ4 (generalisation and losses). *Generalisation is poor and the metric choice is the key.* Models do not transfer out of the box across datasets; off-diagonal ordinal agreement collapses while accuracy stays high and misleading. The in-domain score is also a poor predictor of transfer: the correlation between in-domain and unseen-target QWK is weak for the baselines ($\rho = 0.28$) and slightly negative across the hybrid grid ($\rho = -0.31$), so a model cannot be picked for deployment from its in-domain score alone. Among losses there is no overall winner — cross-entropy gives the best ordinal agreement (QWK) while weighted and ordinal losses recover minority recall and balanced ordinal error — so the loss is a trade-off; notably, the cross-entropy-first order is the one in-domain preference that stays the same on every unseen target, which confirms, after the fact, the choice to fix it before development. QWK together with macro-accuracy and macro-MAE must be the main measure, and training on the datasets pooled together is the single most useful tool for an unknown target.

6.4 Limitations

The findings should be read against several limits, each of which also motivates the future work below.

- **Single-annotator private labels.** The private set was labelled by one annotator (the author) under the public-dataset rubric. Although the model suggestions shown during labelling were optional, the lack of a second annotator means no agreement between annotators can be reported for this set, and the 366-clip size limits how reliable the private-set conclusions are — especially for the ten Highly-Disengage clips, where a single wrong clip moves the recall a lot.
- **Clip-level I3D features.** The I3D features are a single pooled 1024-dimensional vector per clip — shipped ready by CMOSE and extracted by us from the raw video for DAiSEE and the private set — so the I3D "stream" is an MLP over that global motion/appearance summary rather than a temporal encoder; a per-window I3D sequence might add more and would let an I3D temporal encoder model real dynamics.
- **Small in-domain gain and a flat design space.** Most of the 243-configuration grid doesn't care which encoder it gets, and the best in-domain gain over the

baseline is small ($+0.068$ QWK). The decomposition is a sound, easy-to-read framework rather than a large jump in performance.

- **Weak DAiSEE signal.** The very low in-domain QWK on DAiSEE meant it could not be used for architecture development, so the cross-dataset conclusions rest mainly on CMOSE and the private set.
- **Fixed training recipe.** A single seed, learning rate, and encoder size were used throughout; without multi-seed repeats the reported gaps are point estimates rather than confidence intervals, and small differences between nearby configurations should not be over-read.

6.5 Practical Recommendations

For practitioners building engagement-monitoring systems, the study gives three concrete recommendations that do not depend on the specific architecture. First, **report and optimise ordinal agreement**: use QWK, macro-accuracy, and macro-MAE as the headline metrics and treat plain accuracy as secondary, because on these imbalanced ordinal labels accuracy can look healthy while the model has learned nothing useful about disengagement. Second, **train on mixed datasets pooled together** when the deployment data is unknown; pooling was the simplest tool that worked best for the unseen private set and needs no change to the model. Third, **expect and watch rare-class failure**: the most important class, Highly-Disengage, is also the rarest and the hardest to detect, so a deployed system should treat low-confidence disengagement predictions with care and, where possible, leave the call to a human.

6.6 Suggestion for Future Works

- **Targeting the rarest class.** Combine the architecture with imbalance-specific fixes — ordinal losses tuned for Highly-Disengage, minority oversampling such as SMOTE [15], or cost-sensitive decision thresholds — to recover HD recall without trading away Highly-Engage. Because HD is so rare, even small gains here would change the practical value of the system more than further QWK gains on the common classes.
- **Domain adaptation.** The cross-dataset gap calls for explicit unsupervised domain adaptation or domain-invariant representation learning, rather than the plain dataset pooling that already helps; a small amount of labelled target data could be used to calibrate the model before deployment.
- **Learned, not listed, group encoders.** Replace the discrete 243-configuration search with a differentiable architecture search or a gating mechanism that

picks per-group temporal operators end-to-end, so the model finds which encoder suits which group on its own instead of trying them all by hand.

- **Stronger and richer motion features.** Re-extract per-window I3D (or a modern video backbone) so the motion stream carries timing, and add audio/speech features, which CMOSE also provides, as extra semantic streams within the same split-then-fuse framework.
- **Frequency-domain analysis of engagement (exploratory).** A natural extra direction — not done in this thesis and left strictly as future work — is to study the *temporal frequency* of facial signals, under the idea that engaged behaviour is low-frequency and steady while disengaged behaviour is higher-frequency and fidgety, for example through short-time-Fourier-transform features or convolutions over the frequency axis.
- **Stronger evaluation.** Multi-annotator labelling of a larger private set, with reported agreement between annotators, and multi-seed training with confidence intervals would tighten every gap reported here and turn point estimates into statistically grounded claims.

6.7 Closing Remarks

Automatic engagement recognition is appealing exactly because online learning makes the webcam the only window onto the student, but the same setting makes the problem hard: the labels are ordinal, badly imbalanced, subjective, and dataset-specific. This thesis showed that keeping the structure of the facial features — splitting them into behavioural groups and encoding each on its own terms — gives a sound and easy-to-read model that, combined with a motion stream and trained on pooled datasets, generalises best to the most realistic, unseen data. Just as importantly, it showed that the field's usual reliance on accuracy hides a near-total failure to transfer across datasets, and that ordinal agreement metrics show it. Detecting the rarest, most important disengaged behaviour under distribution shift is still open, and is, in the author's view, the most worthwhile next step.

REFERENCE

[1] C.-H. Wu **andothers**, "CMOSE: Comprehensive Multi-Modality Online Student Engagement Dataset with High-Quality Labels," **in***Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* 2024, **pages** 4636–4645. DOI: 10.1109/CVPRW63382.2024.00466

[2] A. Gupta, A. D'Cunha, K. Awasthi **and** V. Balasubramanian, "DAiSEE: Towards User Engagement Recognition in the Wild," *arXiv preprint arXiv:1609.01885*, 2016.

[3] T. Baltrušaitis, A. Zadeh, Y. C. Lim **and** L.-P. Morency, "OpenFace 2.0: Facial Behavior Analysis Toolkit," **in***13th IEEE International Conference on Automatic Face and Gesture Recognition (FG)* 2018, **pages** 59–66. DOI: 10.1109/FG.2018.00019

[4] O. Çopur, M. Nakip, S. Scardapane **and** J. Slowack, "Engagement Detection with Multi-Task Training in E-Learning Environments," **in***Image Analysis and Processing – ICIAP 2022* **jourser** Lecture Notes in Computer Science, **volume** 13233, Springer, 2022, **pages** 411–422. DOI: 10.1007/978-3-031-06433-3_35

[5] S. Malekshahi, J. M. Kheyridoost **and** O. Fatemi, "A General Model for Detecting Learner Engagement: Implementation and Evaluation," *arXiv preprint arXiv:2405.04251*, 2024.

[6] F. M. Shiri, E. Ahmadi, M. Rezaee **and** T. Perumal, "Detection of Student Engagement in E-Learning Environments Using EfficientNetV2-L Together with RNN-Based Models," *Journal on Artificial Intelligence*, **jourvol** 6, **number** 1, **pages** 85–103, 2024. DOI: 10.32604/jai.2024.048911

[7] L. Zhang, X. Zheng, X. Chen **and** L. Cui, "Three-Dimensional View Relationship-Based Context-Aware Emotion Recognition," *IEEE Transactions on Neural Networks and Learning Systems*, **jourvol** 36, **number** 7, **pages** 13 567–13 578, 2025. DOI: 10.1109/TNNLS.2024.3476249

[8] I. Qarbal, N. Sael **and** S. Ouahabi, "Student's Engagement Detection Based on Computer Vision: A Systematic Literature Review," *IEEE Access*, **jourvol** 13, **pages** 140 519–140 545, 2025. DOI: 10.1109/ACCESS.2025.3596885

[9] M. M. Santoni, T. Basaruddin **and** K. Junus, "Convolutional Neural Network Model based Students' Engagement Detection in Imbalanced DAiSEE Dataset," *International Journal of Advanced Computer Science and Applications (IJACSA)*, **jourvol** 14, **number** 3, **pages** 617–626, 2023. DOI: 10.14569/IJACSA.2023.0140371

[10] M. Tan **and** Q. V. Le, "EfficientNetV2: Smaller Models and Faster Training," **in***Proceedings of the 38th International Conference on Machine Learning (ICML)* **jourser** Proceedings of Machine Learning Research, **volume** 139, 2021, **pages** 10 096–10 106.

[11] J. Carreira **and** A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset," **in***Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2017, **pages** 4724–4733. DOI: `10.1109/CVPR.2017.502`

[12] S. Bai, J. Z. Kolter **and** V. Koltun, "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[13] S. Hochreiter **and** J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, **jourvol** 9, **number** 8, **pages** 1735–1780, 1997. DOI: `10.1162/neco.1997.9.8.1735`

[14] A. Vaswani **andothers**, "Attention Is All You Need," **in***Advances in Neural Information Processing Systems (NeurIPS)* 2017, **pages** 5998–6008.

[15] N. V. Chawla, K. W. Bowyer, L. O. Hall **and** W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, **jourvol** 16, **pages** 321–357, 2002. DOI: `10.1613/jair.953`

[16] D. P. Kingma **and** J. Ba, "Adam: A Method for Stochastic Optimization," **in***International Conference on Learning Representations (ICLR)* 2015.

[17] J. Cohen, "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, **jourvol** 20, **number** 1, **pages** 37–46, 1960. DOI: `10.1177/001316446002000104`

[18] C. Spearman, "The Proof and Measurement of Association between Two Things," *The American Journal of Psychology*, **jourvol** 15, **number** 1, **pages** 72–101, 1904. DOI: `10.2307/1412159`

[19] M. G. Kendall, "A New Measure of Rank Correlation," *Biometrika*, **jourvol** 30, **number** 1–2, **pages** 81–93, 1938. DOI: `10.1093/biomet/30.1-2.81`